Marketing Analytics

Chapter 1 · Marketing Analytics as Decision-Making

Nils Wlömert
Vienna University of Economics and Business (WU)

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Who I am…

Contact

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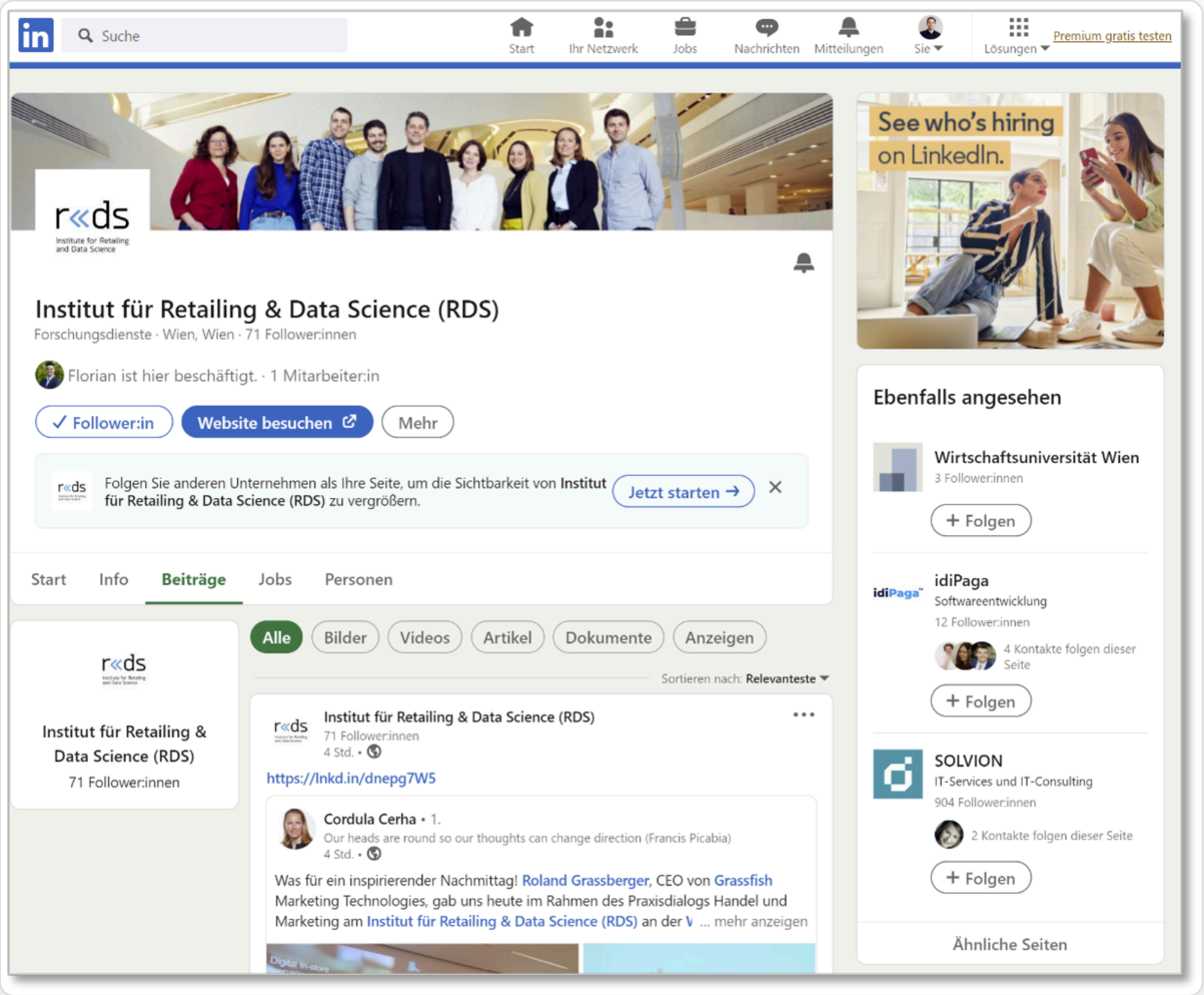
- **Professor** of Marketing, Vienna University of Economics and Business, Institute for Retailing & Data Science, since 2021
- **Assistant Professor** of Marketing, Vienna University of Economics and Business, Institute for Interactive Marketing & Social Media, 2015–2020
- **Ph.D.** in Marketing, University of Hamburg, 2010–2014
- **Dissertation topic:** “Information Technology and Online Content Distribution”
- **Universal Music Group:** Business Analyst & Digital Marketing Manager, 2008–2010
- **Research interests:** multi-channel distribution, online content distribution, digital business models, marketing modeling, e-commerce policies, information goods, targeting & consumer privacy …

Institut für Retailing & Data Science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Cordula Cerha • 1.
Our heads are round so our thoughts can change direction (Francis Picabia)
4 Std. •

Was für ein inspirierender Nachmittag! Roland Grassberger, CEO von Grassfish Marketing Technologies, gab uns heute im Rahmen des Praxisdialogs Handel und Marketing am Institut für Retailing & Data Science (RDS) an der W ... mehr anzeigen

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Contacts

Lecture & group project

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Assignments

Anna Di Donato
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Consultation hours: Fridays 3:00PM - 5:00PM (by appointment)

> **Before you write an email:** course site → AI tools / forum on Canvas → ask in class. If you have the question, so do five others.

Before we start: who are you?

Go to **menti.com**
and enter code **2171 2732**

- Your background & prior R experience
- What you want to get out of this course

Course organization

How this course works

Weekly sessions & self-study

- **Weekly in-person sessions** (see schedule) — concepts, live analysis, discussion, and lots of time for questions
- **Self-study *after* each session:** the course book on the website consolidates the session — chapter text, commented code, and a learning check; slides, code & PDF handouts linked from every chapter
- Each session opens with a short **paper quiz** on the previous chapter *(counts toward participation)*
- **Before session 2:** complete the self-paced **R onboarding module** (install R/RStudio)

Applying the knowledge

- **Three individual assignments** transfer the methods to new business settings — reproducible reports (.Rmd), submitted via Canvas, short in-class debrief *(details in a moment)*
- **Group project:** design & conduct a real market research project — two coaching sessions, results submitted as a **pre-recorded video presentation**

Getting help

- **Forum on Canvas** / **AI-Tools** first; then coaching sessions & in-person sessions

Schedule — Group A (Mondays)

#	Date	Time	Room	Topic
1	Mon, Oct 5	13:00–16:00	TC.4.03	Questions, decisions & data
2	Mon, Oct 12	13:00–17:00	D3.0.233	Describing data
3	Mon, Oct 19	13:00–17:00	D3.0.233	Uncertainty & Experiments I
4	Tue, Oct 27	13:00–16:00	TC.5.05	Experiments II
5	Mon, Nov 2	13:00–16:00	TC.4.03	Observational data I
6	Mon, Nov 9	13:00–16:00	TC.4.03	Observational data II
7	Mon, Nov 16	13:00–16:00	TC.4.03	Mechanisms & heterogeneity
8	Mon, Nov 23	13:00–16:00	TC.4.03	Survey design

Assignments — Group A (Mondays)

#	Topic	Hand-out	Submission	Debrief
A1	Data handling & description in R	Mon, Oct 12	Mon, Oct 26	Tue, Oct 27
A2	Analyzing experiments	Tue, Oct 27	Sun, Nov 8	Mon, Nov 9
A3	Analyzing observational data	Mon, Nov 9	Sun, Nov 22	Mon, Nov 23

ASSIGNMENTS ON SYLLABUS ARE
CLOSER THAN THEY APPEAR

- Submitted as a **reproducible report** (.Rmd → HTML) via Canvas — **deadline: the day before the debrief session**
- Every student receives an **individual dataset** — results and conclusions differ **by design**
- **Debrief** = 3 short paper questions about *your* results at the start of the next session — **report 7 + debrief 3 points**; your lowest debrief score is replaced by the average of the other two

Schedule — Group B (Thursdays)

#	Date	Time	Room	Topic
1	Thu, Oct 8	13:00–16:00	TC.5.03	Questions, decisions & data
2	Thu, Oct 15	13:30–17:30	TC.3.03	Describing data
3	Thu, Oct 22	13:00–17:00	TC.5.15	Uncertainty & Experiments I
4	Thu, Oct 29	13:00–16:00	TC.5.03	Experiments II
5	Thu, Nov 5	13:00–16:00	TC.5.03	Observational data I
6	Thu, Nov 12	13:00–16:00	TC.5.03	Observational data II
7	Thu, Nov 19	13:00–16:00	TC.5.03	Mechanisms & heterogeneity
8	Thu, Nov 26	13:30–16:30	TC.3.03	Survey design

Assignments — Group B (Thursdays)

#	Topic	Hand-out	Submission	Debrief
A1	Data handling & description in R	Thu, Oct 15	Wed, Oct 28	Thu, Oct 29
A2	Analyzing experiments	Thu, Oct 29	Wed, Nov 11	Thu, Nov 12
A3	Analyzing observational data	Thu, Nov 12	Wed, Nov 25	Thu, Nov 26

ASSIGNMENTS ON SYLLABUS ARE
CLOSER THAN THEY APPEAR

- Submitted as a **reproducible report** (.Rmd → HTML) via Canvas — **deadline: the day before the debrief session**
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- **Debrief** = 3 short paper questions about *your* results at the start of the next session — **report 7 + debrief 3 points**; your lowest debrief score is replaced by the average of the other two

Group project & final exam — key dates

Milestone	Group A (Mondays)	Group B (Thursdays)
Coaching I — questionnaire design *(online)*	Mon, Nov 30, 13:00–17:00	Thu, Dec 3, 14:00–18:00
Field your survey	December	December
Coaching II — data analysis *(online)*	Thu, Jan 7, 13:00–17:00	Fri, Jan 8, 14:00–18:00
Submission: video presentation	Wed, Jan 20	Wed, Jan 20
Submission: peer review of one other group’s video	Mon, Jan 25	Mon, Jan 25

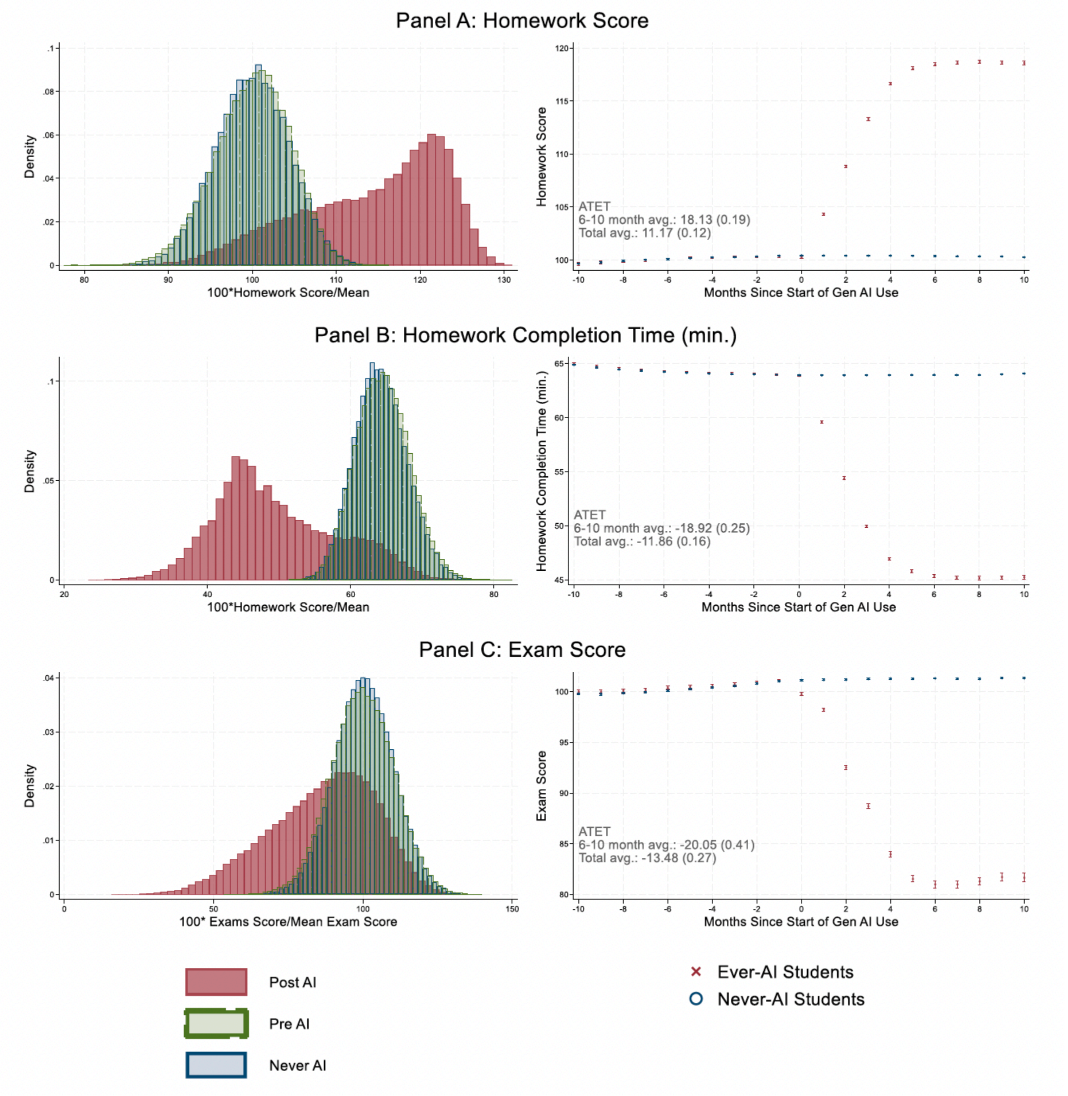
Final exam (both groups): Tue, **Dec 15**, 10:00–12:00, D5.0.001 — *paper & pencil, open book (slides), in person*

Our AI policy

AI assistants are welcome in the assignments — they may support your work at every step, but not do it for you: a submission without individual contribution earns no credit, disclosed or not. You must be able to **explain everything you submit** (that is what the debrief checks), and each report ends with a short **AI-use disclosure**. Honest disclosure is never penalized; concealing substantial AI use is. The final exam is the unassisted layer: paper & pencil, in person.

"How much AI is allowed?" — As much as you like,
provided you can stand here and explain what was done and why.

Use it — but don't let it study for you



- **26,811 Chinese students**, 30-month panel: AI adoption → homework **+18%**, time **−30%** — but exam scores **−20%**; entrance exams **−18/−24%** after two years
- Driven by the **~80% who outsource**; students who used AI but spent **the same time** on homework: **no learning loss**

Sources: Strömberg, Lei & Wu (2026); Autor et al. (2026, NBER WP 35720); Bastani et al. (2025, PNAS); Kestin et al. (2025). Figures: Strömberg et al. (2026), Fig. 2 — red ×: AI adopters, blue ○: never-AI students.

Grading

Component	Weight
Market research **group project** (design → data collection → analysis → **video presentation**)	30%
Individual **take-home assignments** (3 × reproducible report + **in-class debrief**; each 7 + 3 points)	30%
Final exam (concepts & methods — no coding · **course slides may be used**, paper & pencil, in person)	30%
Class participation (weekly quizzes · active contributions in class & forum)	10%

- Pass threshold I: **60%** weighted total
- Pass threshold II: you need to be present **80%** of the in-class time
- Group project: **peer assessment** + **contribution & AI-use statement** enter the individual grade
- Participation = **quizzes** (up to 6 points, best 6 of 7) **+ active contributions** (up to 6 points, max. 1 per session, incl. forum) — **capped at 10**: strengths in one can compensate for the other
- Assignment dates: see slides · all details on the course site

Grading scheme

Overall percentage	Grade
90–100%	1 (excellent)
80–89%	2 (good)
70–79%	3 (satisfactory)
60–69%	4 (sufficient)
< 60%	5 (fail)

> To pass, your **weighted final grade** must reach at least **60%** — across all components combined; there is no minimum per component.

Course materials

- **Main reference: the course website** — full chapters with explanations, commented R code, and learning checks; **slides & PDF handouts** linked from every chapter
- Everything is **free & always current** — no textbook purchase required
- **Optional further reading** linked on the site
- **Canvas**: assignment hand-outs & submissions, forum, announcements
- Example exams: provided on Canvas **before the exam**

The group project in one sentence

You will **design a survey containing a real randomized experiment**, field it, analyze the data, and present the results — everything this course teaches, built by you.

(Details follow — and at the end of today's session you will see why "randomized" is the most important word in that sentence.)

The group project: two parts

Part 1 · Questionnaire design & data collection

- Pick a topic: **14 proposed topics** (with starter literature) — or suggest your own
- Design a questionnaire (≈ **30 questions**, ~5 per member) in **Qualtrics** — free account provided
- Your design contains a **real experiment**: participants randomly assigned to conditions
- **Coaching I:** present your draft (research problem, design, measurement, planned analyses) → feedback → revise
- **Field it:** each member recruits ≈ **20 respondents** → ≈ **120 responses** per group

Part 2 · Data analysis & presentation

- Import & clean your Qualtrics data in R
- Analyze: descriptives → your **experimental comparison** → heterogeneity (for whom does it work?)
- **Coaching II:** work through your analysis with us *(online)*
- Submit a **.zip**: pre-recorded **video presentation** + data + reproducible R code + slides
- Afterwards: **peer review** — each of you reviews **one** other group's video (structured form)

Individual grades: peer assessment within the group + a short **contribution & AI-use statement** + your **peer review** of one other video (2 of 30 points).

Group project: what makes it good

- A **precise research question** with managerial relevance — and hypotheses stated *before* data collection
- A clean **experimental manipulation**: everything constant except the treatment; randomizer placed correctly
- **Keep it simple**: 2 conditions are sufficient (your sample size will be limited)
- **Deliberate measurement:** a dependent variable with real variation (e.g., willingness to pay)
- Incorporate **moderators** to analyse heterogeneous treatment responses
- Incorporate **mediators** to understand the process
- Analyses **matched to the design** — justified test choice, honest uncertainty, assumptions checked
- A presentation that **tells the decision-maker what to do** — not a parade of charts

Full guidelines, topic list, Qualtrics templates & how-tos: **“Group project” section on the course site**.

Course goals

After taking this course, you should…

- …know why analytical skills have become **career-defining** for marketers — and which skills stay scarce when AI does the mechanics
- …know **what data can and cannot tell you** — and recognize which question you are actually asking: *describe, predict, explain, decide*
- …know **when to use which method** — and which assumptions the answer rests on
- …be able to **implement the methods in R** — with AI assistants as accelerators, not replacements
- …be able to **judge analyses made by others** — a colleague's slide deck, a platform dashboard, an AI-generated report
- …be able to **conduct your own research project** — from question to data to recommendation
- …no longer be **afraid of numbers & statistics** (in case you were)
- …feel the **desire to learn more** about analytics

Today's session

Part A — Why we are here

1.1 Motivation: data and AI in marketing
1.2 Case study: search advertising at eBay

Part B — The framework of the course

1.3 Types of research questions
1.4 Correlation and causation
1.5 Variables and their roles
1.6 Randomization
1.7 Quasi-experiments

Part C — Making it work

1.8 From business problem to research design
1.9 The analytics workflow

> **One promise for today:** no R output, no tests, and just one simple formula. Concepts and pictures — the machinery follows in later sessions.

1.1 Motivation: data and AI in marketing

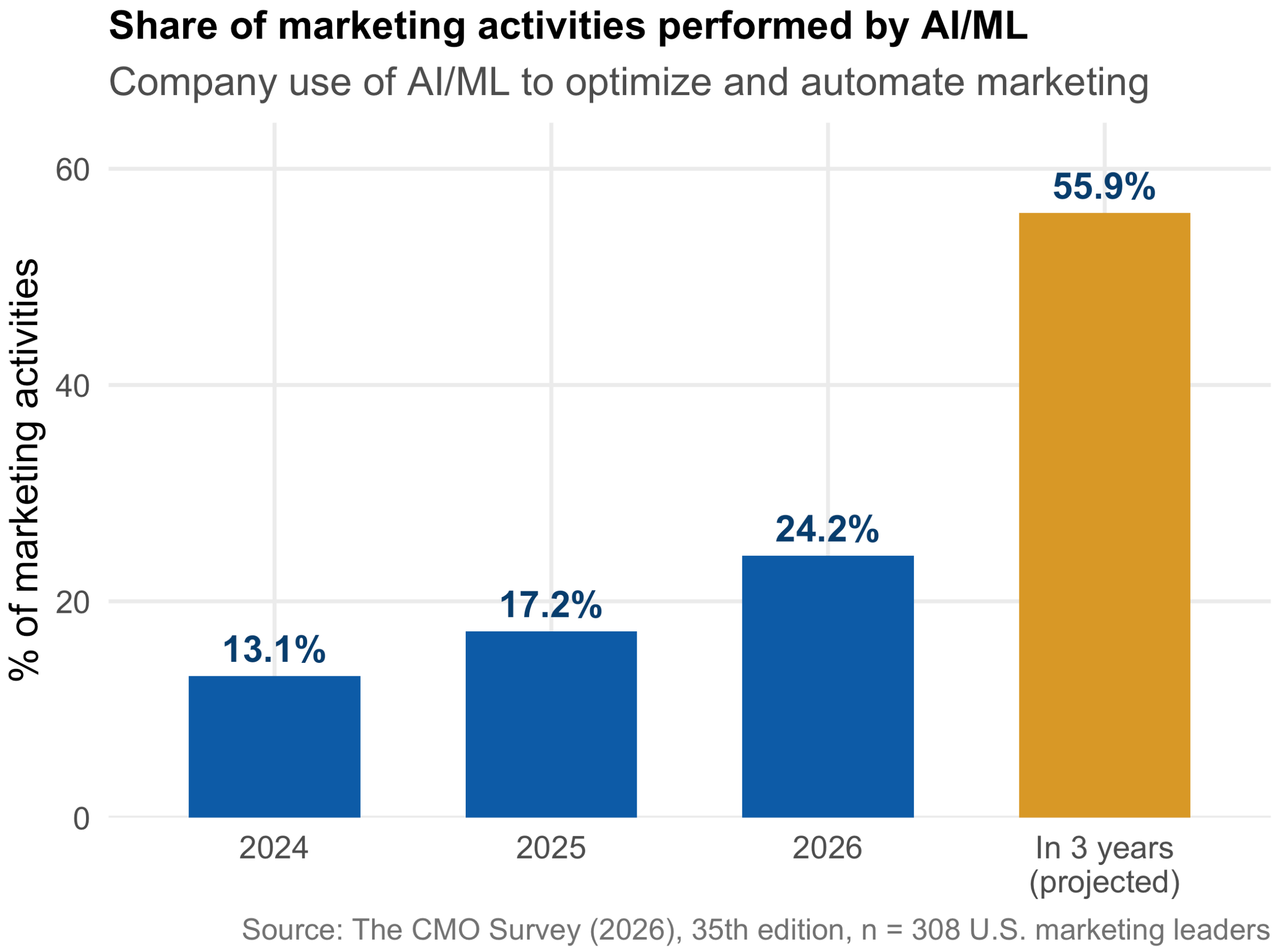
“I don’t know which half”

> “Half the money I spend on advertising is wasted; the trouble is I don’t know which half.”
>
> — John Wanamaker (1838–1922), department store pioneer, Philadelphia

More than 100 years later, **do we know** today which half of the advertising budget is wasted? What **data** would you need to find out?

Marketing is reorganizing around AI and data

- AI already performs **about a quarter** of marketing activities — nearly **doubled** in two years
- Generative AI alone: from 7.0% to 22.4% of activities (**+220%** in two years)
- Marketing leaders expect **more than half** of marketing work to be performed by AI **within three years**



Source: The CMO Survey (2026), 35th edition, n = 308 U.S. marketing leaders

… while budgets are flat: every euro must justify itself

1.7%

growth of total marketing spending — the **weakest since 2021**

15.3%

of marketing budgets now go to **AI initiatives**

78.2%

name **“higher ROI per marketing dollar”** as the #1 reason to build marketing capabilities

> “CMOs are being asked to deliver growth, efficiency and transformation **without meaningful budget expansion**. Those who succeed will make deliberate, **data-driven trade-offs**.”
>
> — Gartner CMO Spend Survey (2026)

Sources: The CMO Survey (2026); Gartner CMO Spend Survey (2026), n = 401 CMOs, North America & Europe.

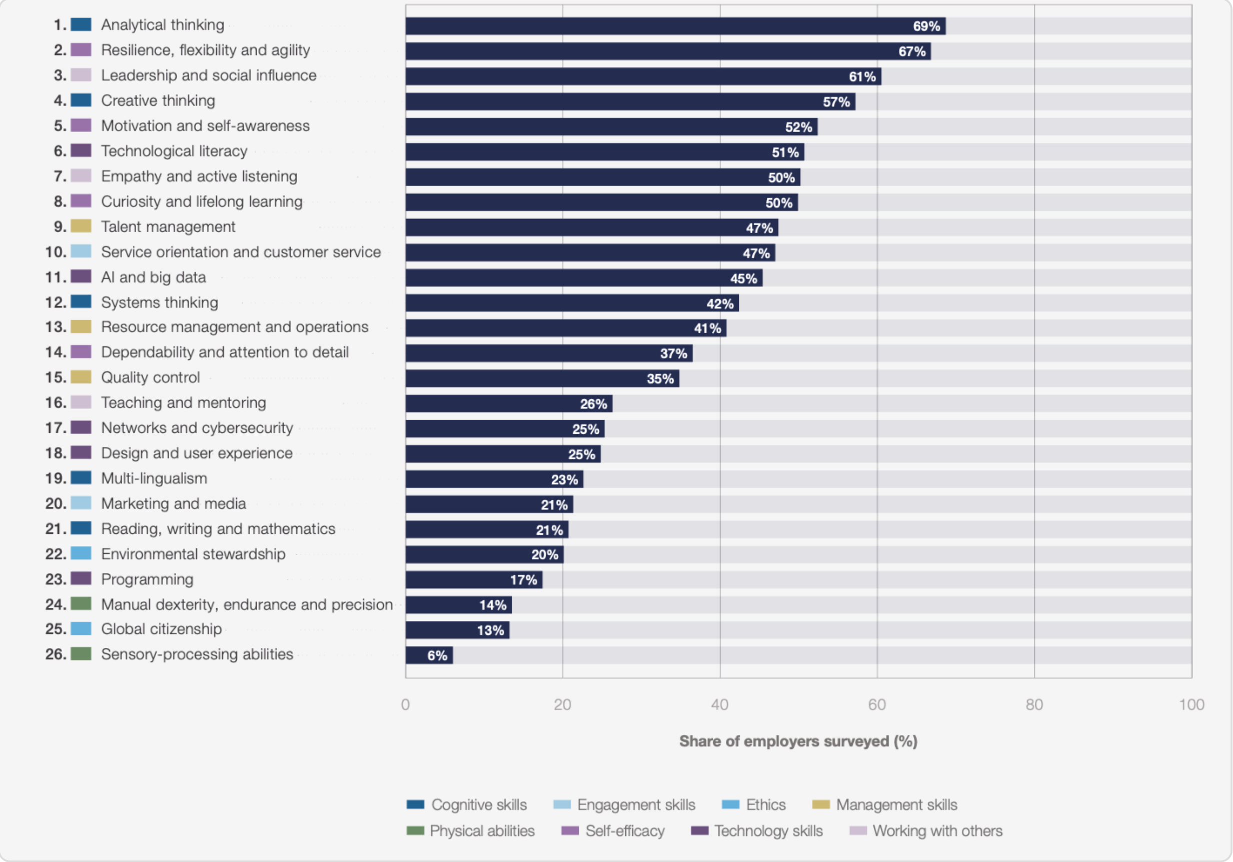
The scarce skill is judgment

The ambition–readiness gap (Gartner 2026):

- **70%** of CMOs: becoming an **AI leader** is a critical goal
- … but only **30%** are ready to **scale** AI capabilities

The labor market (WEF Future of Jobs 2025):

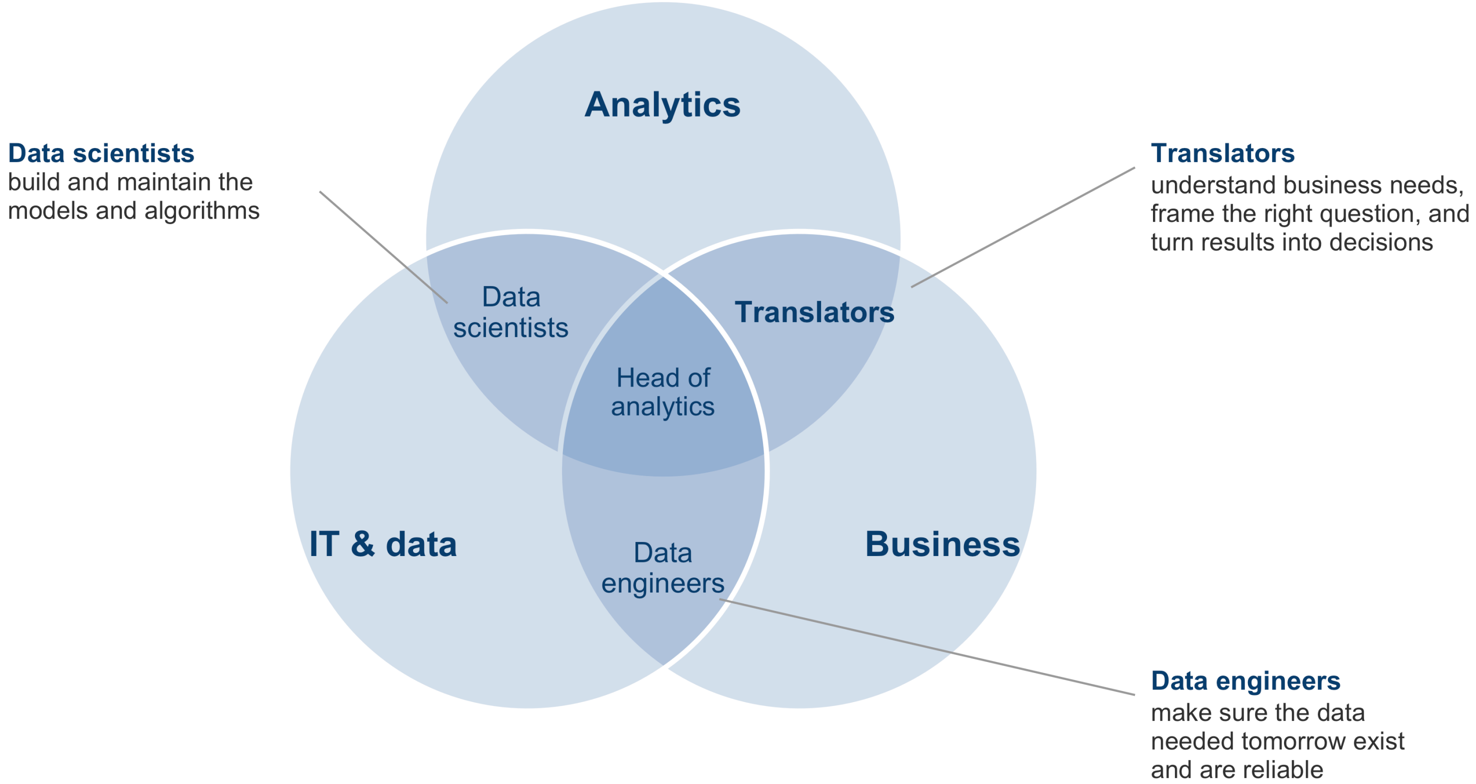
- **Analytical thinking = the #1 core skill** — 7 of 10 employers consider it essential
- **AI & big data = the fastest-growing skill** for 2025–2030



Source: World Economic Forum, Future of Jobs Report 2025

> **The mechanical layer of marketing is being automated. What remains scarce is judgment:** knowing which analysis answers which managerial question, and whether the resulting number can be trusted.

Who turns analytics into decisions?


Analytics
IT & data
Business
Data scientists
Translators
Head of analytics
Data engineers

Data scientists
build and maintain the models and algorithms

Translators
understand business needs, frame the right question, and turn results into decisions

Data engineers
make sure the data needed tomorrow exist and are reliable

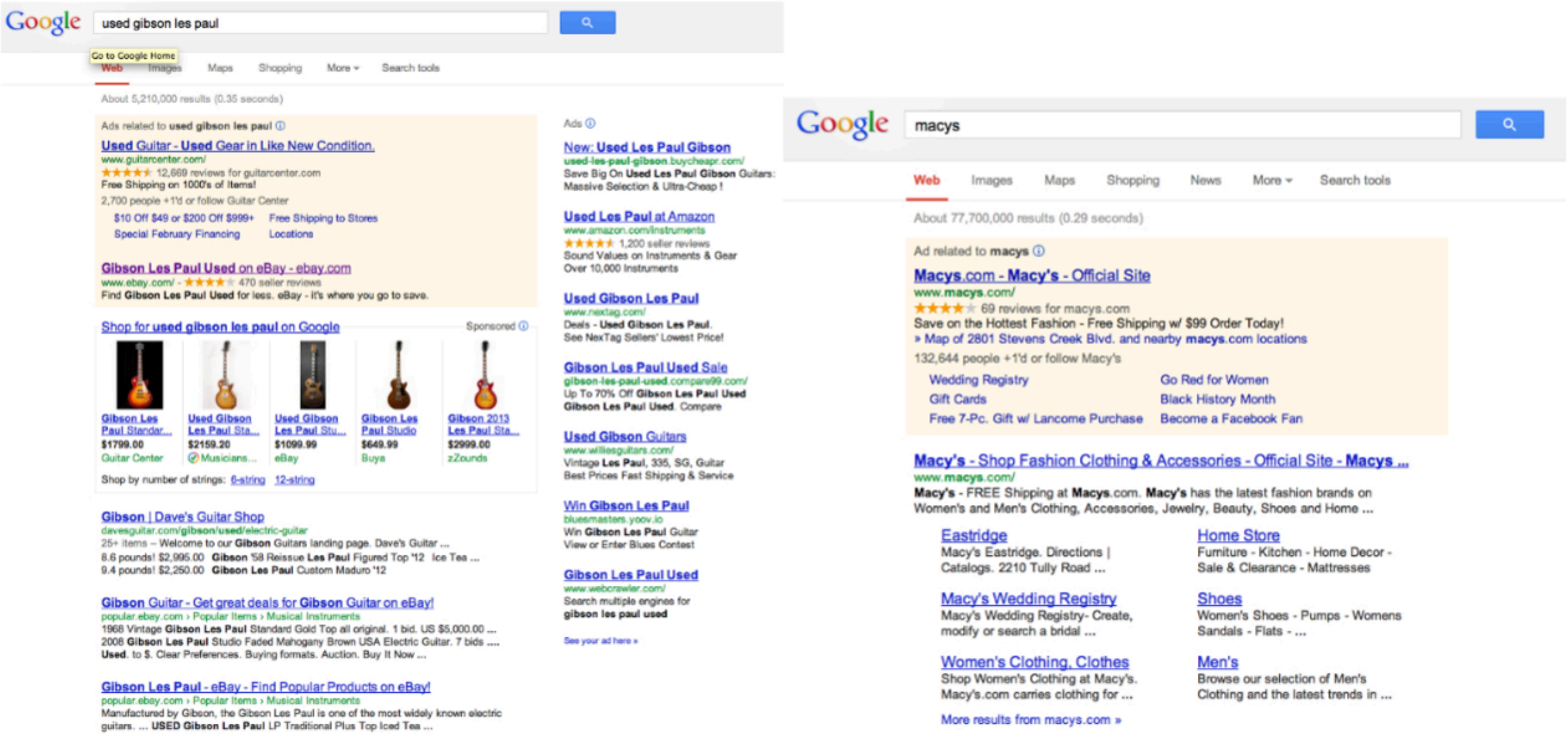
> The **translator** role — framing the business question, choosing the right analysis, judging what the result does and does not show — is exactly what this course trains.

Own illustration; concept after Henke, Levine & McInerney (2018), “Analytics translator: The new must-have role”, *McKinsey Quarterly* (link).

1.2 Case study: search advertising at eBay

eBay, 2012: Does paid search advertising pay off?

Figure 1: Google Ad Examples



About 5,210,000 results (0.35 seconds)

Ads related to **used gibson les paul**

Used Guitar - **Used** Gear in Like New Condition.
www.guitarcenter.com/
★★★★★ 12,669 reviews for guitarcenter.com
Free Shipping on 1000's of Items!
2,700 people +1'd or follow Guitar Center
$10 Off $49 or $200 Off $999+ Free Shipping to Stores
Special February Financing Locations

Gibson Les Paul Used on eBay - ebay.com
www.ebay.com/ - ★★★★ 470 seller reviews
Find **Gibson Les Paul Used** for less. eBay - it's where you go to save.

Shop for **used gibson les paul** on Google — Sponsored

Gibson Les Paul Standar...	Used Gibson Les Paul Sta...	Used Gibson Les Paul Stu...	Gibson Les Paul Studio	Gibson 2013 Les Paul Sta...
$1799.00	$2159.20	$1099.99	$649.99	$2999.00
Guitar Center	Musicians...	eBay	Buya	zZounds

Shop by number of strings: 6-string 12-string

Gibson | Dave's Guitar Shop
davesguitar.com/**gibson**/**used**/electric-guitar
25+ items – Welcome to our **Gibson** Guitars landing page. Dave's Guitar ...
8.6 pounds! $2,995.00 **Gibson** '58 Reissue **Les Paul** Figured Top '12 Ice Tea ...
9.4 pounds! $2,250.00 **Gibson Les Paul** Custom Maduro '12

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1968 Vintage **Gibson Les Paul** Standard Gold Top all original. 1 bid. US $5,000.00 ...
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Used. to $. Clear Preferences. Buying formats. Auction. Buy It Now ...

Gibson Les Paul - eBay - Find Popular Products on eBay!
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Manufactured by Gibson, the Gibson Les Paul is one of the most widely known electric guitars. ... **USED Gibson Les Paul** LP Traditional Plus Top Iced Tea ...

Ads

New: **Used Les Paul Gibson**
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Save Big On **Used Les Paul Gibson** Guitars: Massive Selection & Ultra-Cheap !

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★★★★★ 1,200 seller reviews
Sound Values on Instruments & Gear
Over 10,000 Instruments

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Deals - **Used Gibson Les Paul**.
See NexTag Sellers' Lowest Price!

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Up To 70% Off **Gibson Les Paul Used**
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Best Prices Fast Shipping & Service

Win **Gibson Les Paul**
bluesmasters.yoov.io
Win **Gibson Les Paul** Guitar
View or Enter Blues Contest

Gibson Les Paul Used
www.webcrawler.com/
Search multiple engines for
gibson les paul used

See your ad here »

(a) Used Gibson Les Paul

About 77,700,000 results (0.29 seconds)

Ad related to **macys**

Macys.com - **Macy's** - Official Site
www.**macys**.com/
★★★★ 69 reviews for macys.com
Save on the Hottest Fashion - Free Shipping w/ $99 Order Today!
» Map of 2801 Stevens Creek Blvd. and nearby **macys**.com locations
132,644 people +1'd or follow Macy's
Wedding Registry Go Red for Women
Gift Cards Black History Month
Free 7-Pc. Gift w/ Lancome Purchase Become a Facebook Fan

Macy's - Shop Fashion Clothing & Accessories - Official Site - **Macys ...**
www.**macys**.com/
Macy's - FREE Shipping at **Macys**.com. **Macy's** has the latest fashion brands on Women's and Men's Clothing, Accessories, Jewelry, Beauty, Shoes and Home ...

Eastridge Macy's Eastridge. Directions | Catalogs. 2210 Tully Road ...	Home Store Furniture - Kitchen - Home Decor - Sale & Clearance - Mattresses
Macy's Wedding Registry Macy's Wedding Registry- Create, modify or search a bridal ...	Shoes Women's Shoes - Pumps - Womens Sandals - Flats - ...
Women's Clothing, Clothes Shop Women's Clothing at Macy's. Macy's.com carries clothing for ...	Men's Browse our selection of Men's Clothing and the latest trends in ...

More results from macys.com »

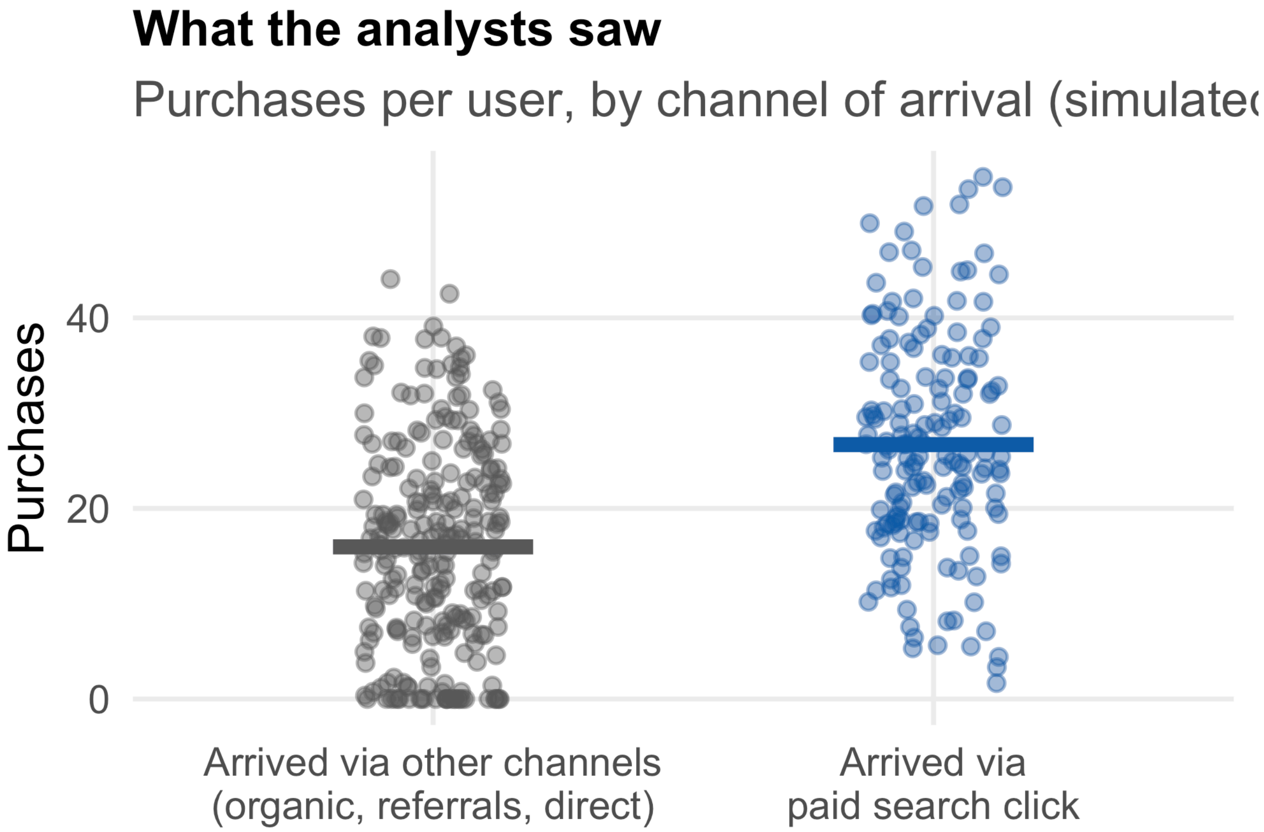
(b) Macys

Blake, Nosko & Tadelis (2015), “Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment”, *Econometrica*, 83(1), 155–174, Fig. 1 — Google results pages for the searches “used gibson les paul” and “macys”.

eBay, 2012: what the dashboards showed

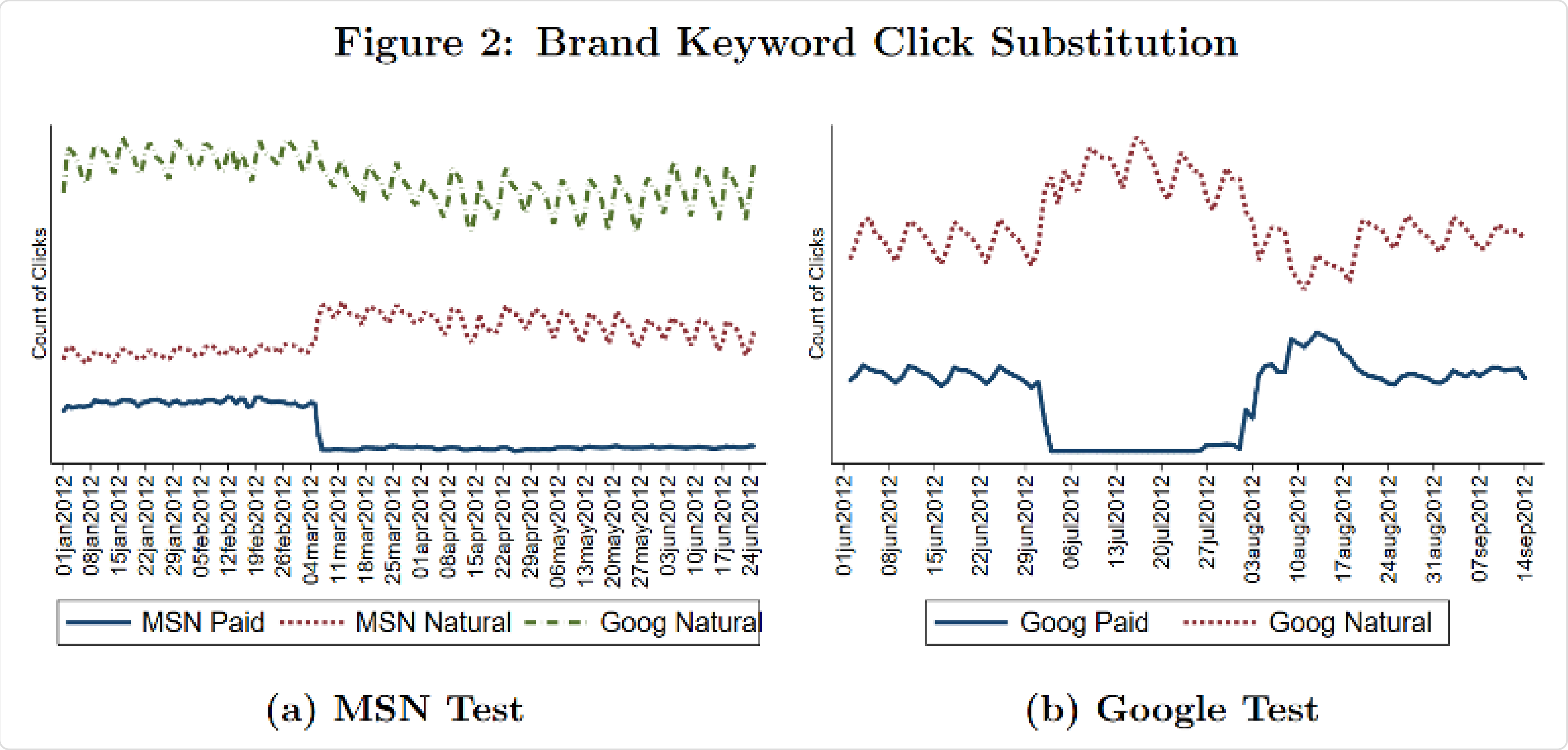
eBay spent ~ **$50M per year** on paid search ads (Google, Bing, Yahoo). Here is how the return on this spending was measured:

1. A user types **“ebay guitar”** into Google and clicks eBay’s **paid ad** at the top of the results page
2. The **tracking parameter** in the URL logs her visit as *arrived via paid search* (with keyword & cost-per-click)
3. All other visitors are logged with a different **referrer** (organic search, other sites, direct) — and serve as the **comparison group**
4. Purchases within the **attribution window** are credited to the arrival channel → the picture on the right, and with it: **attributed revenue ÷ ad spend**, per keyword, updated daily



Discuss (5 min): As eBay’s CMO, would you sign off on next year’s $50M based on this evidence? If yes — what convinces you? If not — **how would you measure the payoff instead?**

What happened when the ads were switched off



Paid clicks fell to zero — and **~99.5% of that traffic instantly reappeared via the free organic link**. Total visits to eBay: essentially unchanged.

Blake, Nosko & Tadelis (2015), "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment", *Econometrica*, 83(1), 155–174, Fig. 2 — brand-keyword ads halted on MSN (March 2012) and Google (July–August 2012); solid line: paid clicks, dotted: organic clicks.

Reading the fine print

Brand keywords ("ebay", "ebay shoes") — the platform halts:

- The users clicking the ad were **already on their way to eBay** — the ad intercepted traffic the free organic link would have caught (**~99.5% substitution**)
- Attributed revenue: almost entirely **re-labeled, not created** → true effect of brand ads ≈ **zero**

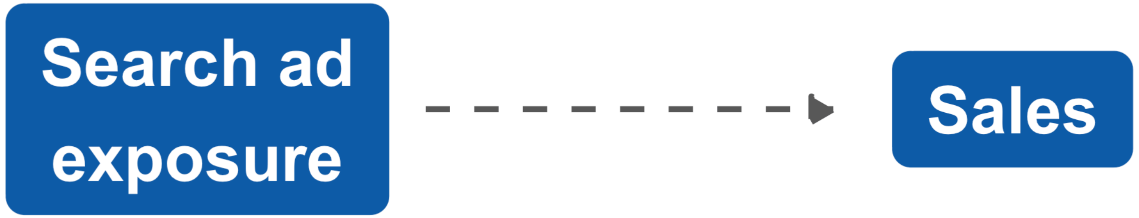
Non-brand keywords ("used guitar") — the regional experiment:

- Ads suspended in a **randomly selected ~30% of US regions for 60 days**; **total sales** compared against the remaining regions — a *real* comparison group, created by design
- Sales moved **only among new and infrequent users**; **frequent shoppers** — most of the clicks and most of the ad cost — were unmoved
- **The customers most likely to click were the customers least influenced** → averaged over everyone, returns on the spending: **negative**

> Smart people, huge data, careful computation — and a conclusion off by tens of millions of dollars. **The problem was not the math. It was the comparison.**

Blake, Nosko & Tadelis (2015), *Econometrica* — details in the course book.

Why the dashboard and the experiment disagree

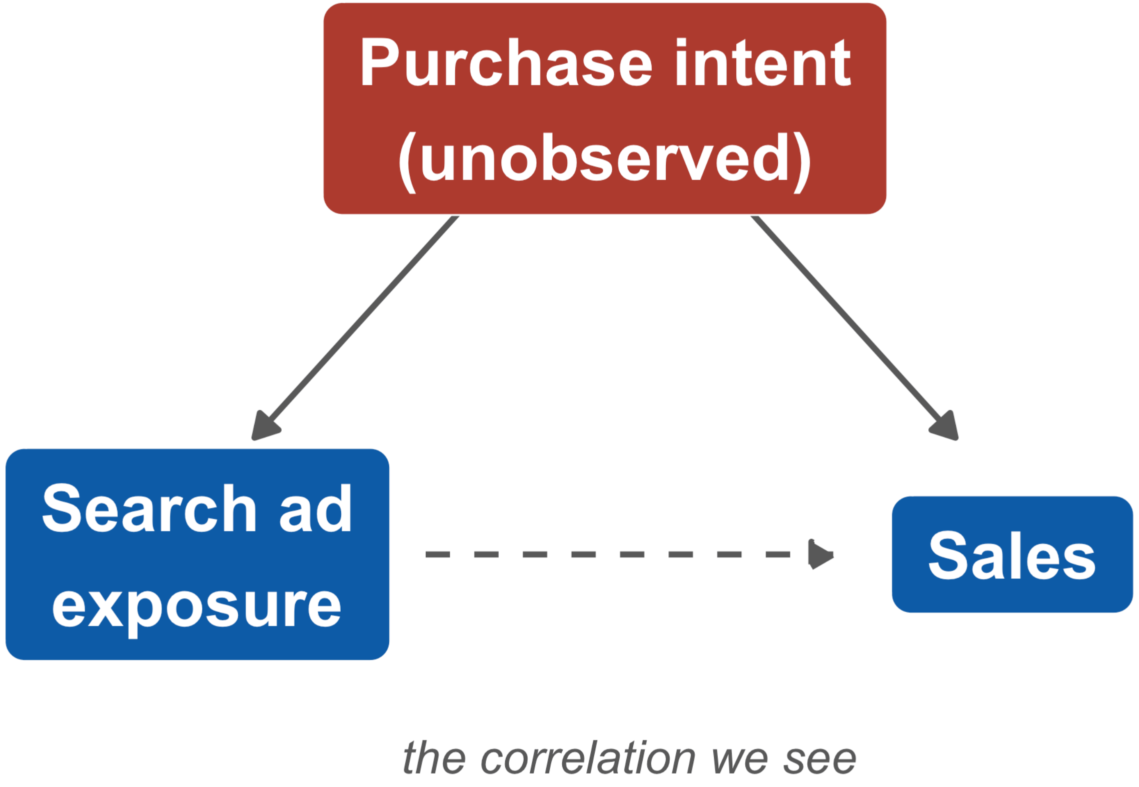


Search ad exposure

Sales

the correlation we see

Why the dashboard and the experiment disagree


Purchase intent (unobserved)
Search ad exposure
Sales
the correlation we see

> **Purchase intent drives both** who sees the ads *and* who buys. The dashboard's correlation mixes the ad effect with the intent difference — and in this case, intent was nearly the whole story.

The comparison that matters

The causal question: *What would eBay's sales have been* ***without*** *the ads — everything else unchanged?*

	What eBay observed	**What the decision needs**
Group A	Users arriving via **paid search**	The same users, **with** ads
Group B	Users arriving via **other channels**	**The same users, without ads**
The difference measures…	Ad effect **+ intent difference**	The ad effect. **Nothing else**

> The needed Group B is a **counterfactual** — a world we can never observe. All of causal inference is about **constructing a credible stand-in for it** — and how well a stand-in can be constructed depends on **how the data came into being**. This one idea organizes the entire course.

1.3 Types of research questions

Every analytics task answers one of four questions

#	Question	Type	Marketing examples
1	**What is going on?**	Descriptive	Segment sizes, churn rate, funnel drop-off, satisfaction tracking
2	**What will happen?**	Predictive	Which customers will churn next quarter? Demand next month?
3	**What happens if we act?**	Causal	Does the discount *cause* extra sales? Did the campaign work?
4	**What should we do?**	Prescriptive	Set the price; allocate budget across channels

> **Prescription requires causation.** To choose an action you must know what each action would *cause* — i.e., compare counterfactuals. Prediction alone is not enough… and confusing 2 with 3 was eBay's mistake.

Four types of questions, four kinds of analytics

	Description	Prediction	Causal inference	Prescription
Typical question	How can our customers be grouped into segments based on their characteristics?	Which of last year's visitors will purchase from our store next month?	Does behavioral targeting *increase* the probability of purchase — and by how much?	Which customers should we target, at what budget, to maximize profit?
What the data must contain	*Features:* user characteristics (age, location, …), products viewed, …	*Inputs* (features) and the *output* to be predicted (purchase next month)	*Outcome*, *treatment*, and — without an experiment — the relevant *confounders*	Causal effect estimates *plus* costs, margins, and constraints
Typical analytics	Cluster analysis, descriptive statistics, visualization	Regression, trees, random forests, neural networks	Randomized experiments; regression, diff-in-diff, IV, RDD	Targeting & uplift models, optimization, simulation

Based on the classification of data-science tasks in Hernán, Hsu & Healy (2019), *Chance* — prescription column added.

Prediction ≠ causation: same data, different questions

“Customers who see our newsletter churn 40% less.”

As a predictive statement ✓

Newsletter readership *predicts* retention — useful for **targeting**: flag non-readers as churn risks.

Valid even if the newsletter does nothing.

As a causal statement ✗ (so far)

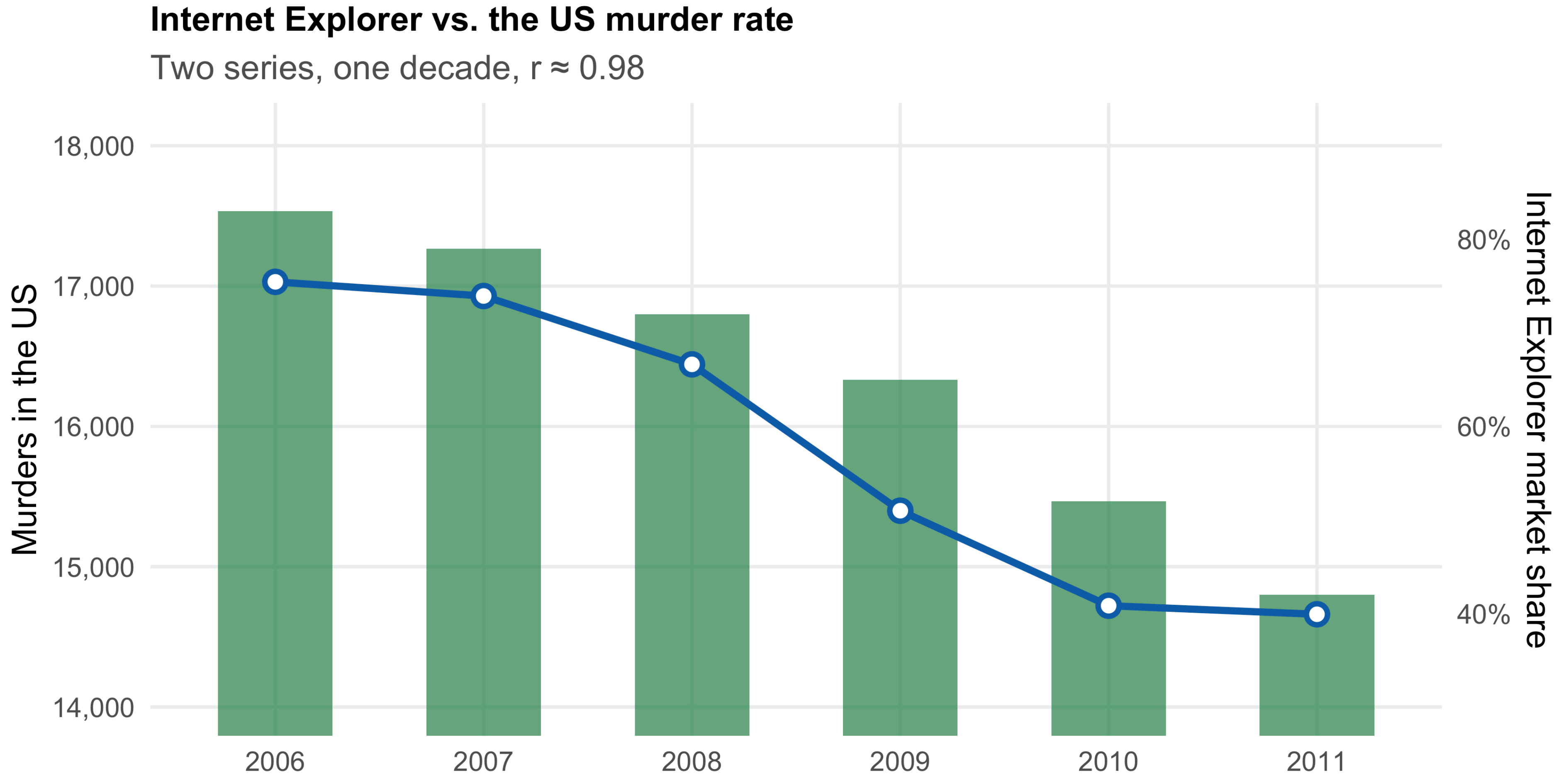
“Send more newsletters → churn drops”? Only if readership *causes* retention — but engaged customers **both** read newsletters **and** stay.

Requires a counterfactual comparison.

Classify these (2 min, neighbor discussion): (a) “Which ad creative will get the highest CTR?” (b) “How large is our Gen-Z segment?” (c) “Should we raise the price by 5%?”

1.4 Correlation and causation

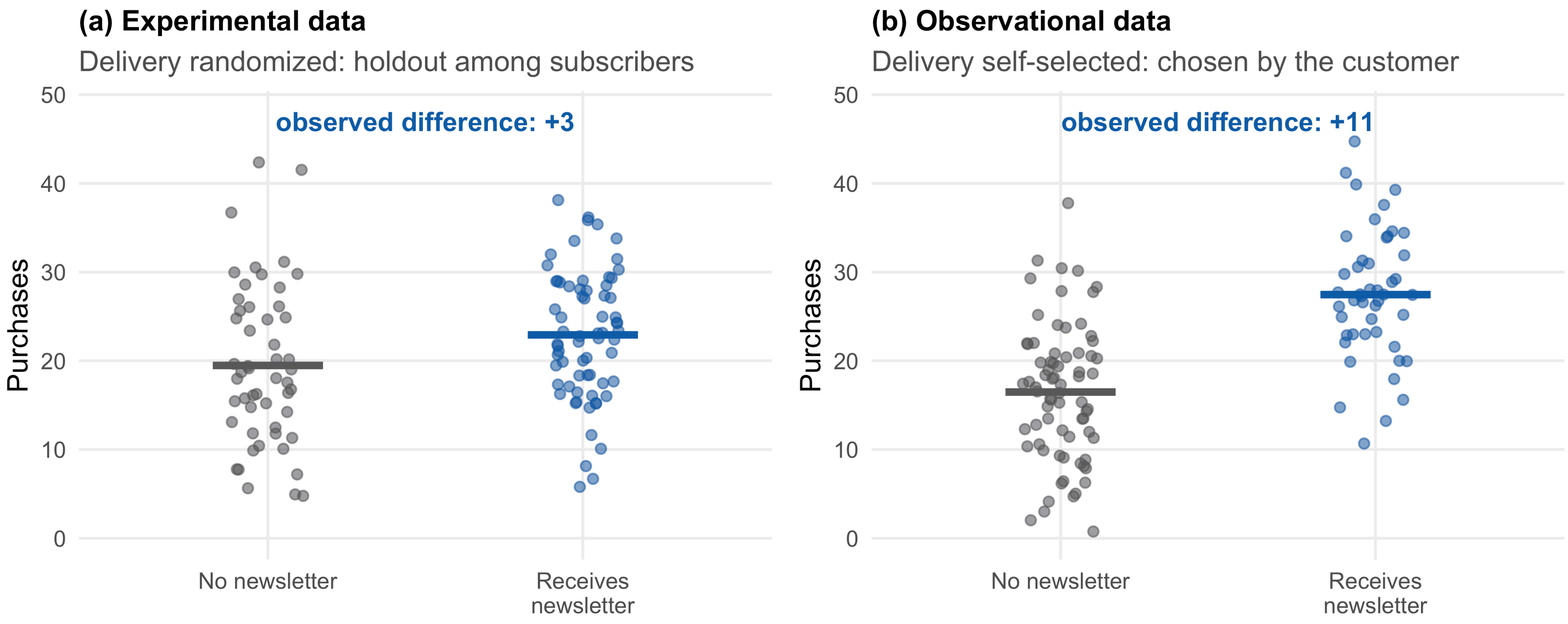
Did Internet Explorer prevent murders?



A **spurious correlation**: two variables move together **without any causal link** — here simply because both **trend in the same direction over time**. Correlation is cheap; causation is not.

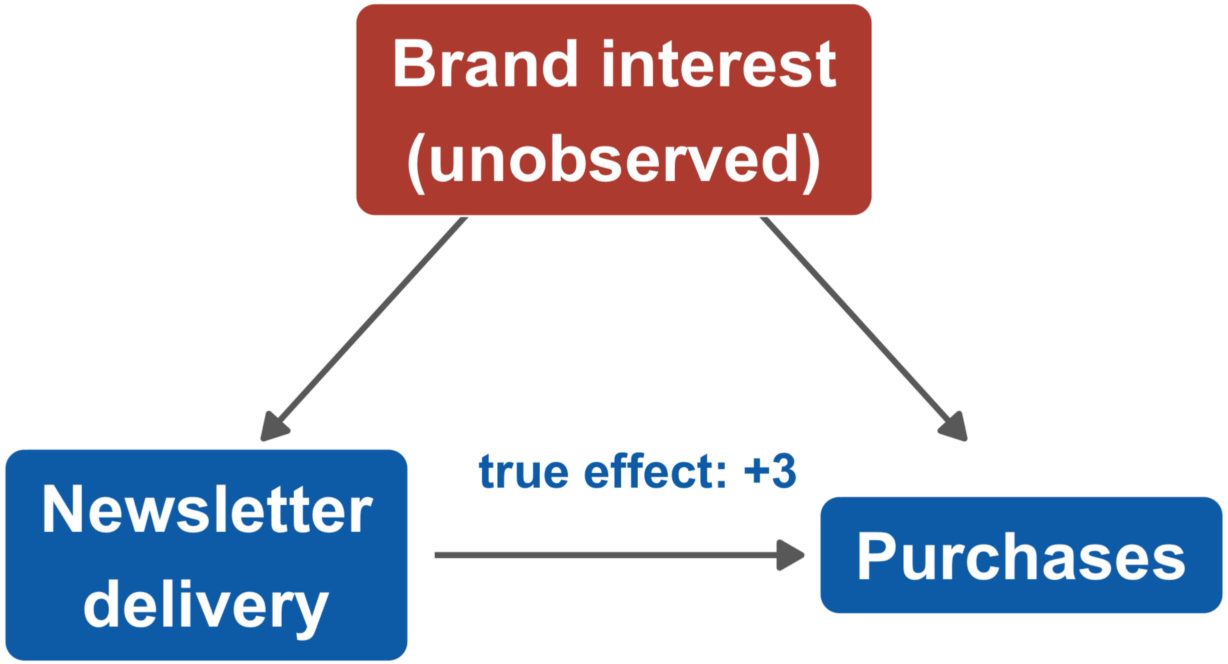
Data: FBI Uniform Crime Reports; StatCounter (approx.). Hundreds more at tylervigen.com/spurious-correlations — chart style after the internet classic.

Two pictures of “X relates to Y”



Same newsletter, same chart — **and the true effect is +3 in both worlds.** Panel (a) recovers it; panel (b) reports **+11**, because customers who subscribe differ from those who don’t *before* any newsletter arrives. Which panel is which is **invisible in the plot**: it lies in **how the data came to be** (the *data-generating process*).

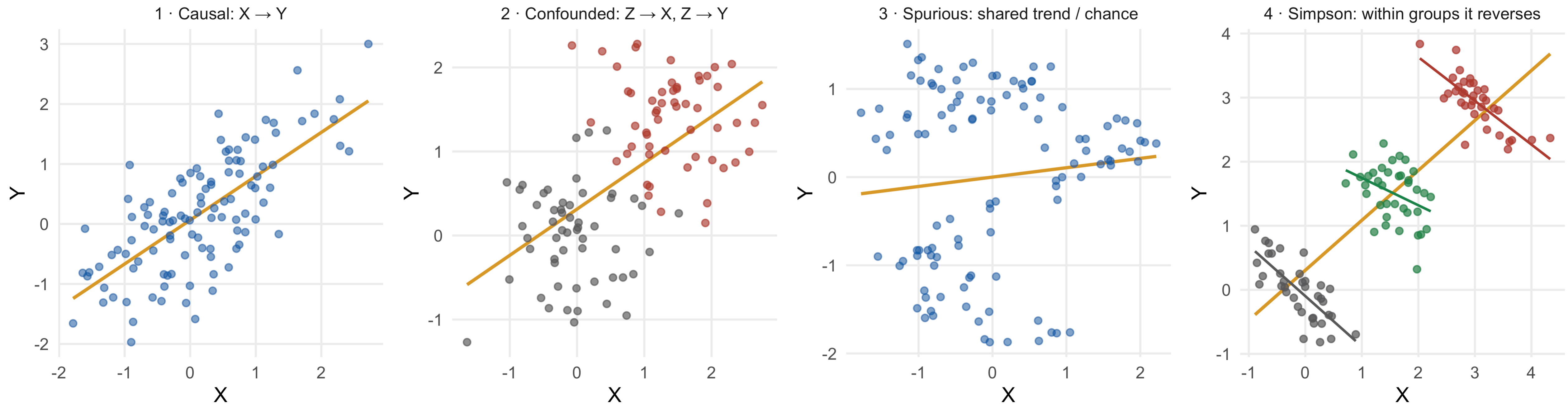
What generates panel (b)


Brand interest (unobserved)
Newsletter delivery
true effect: +3
Purchases

observed in (b): +11 = +3 effect + ~8 selection

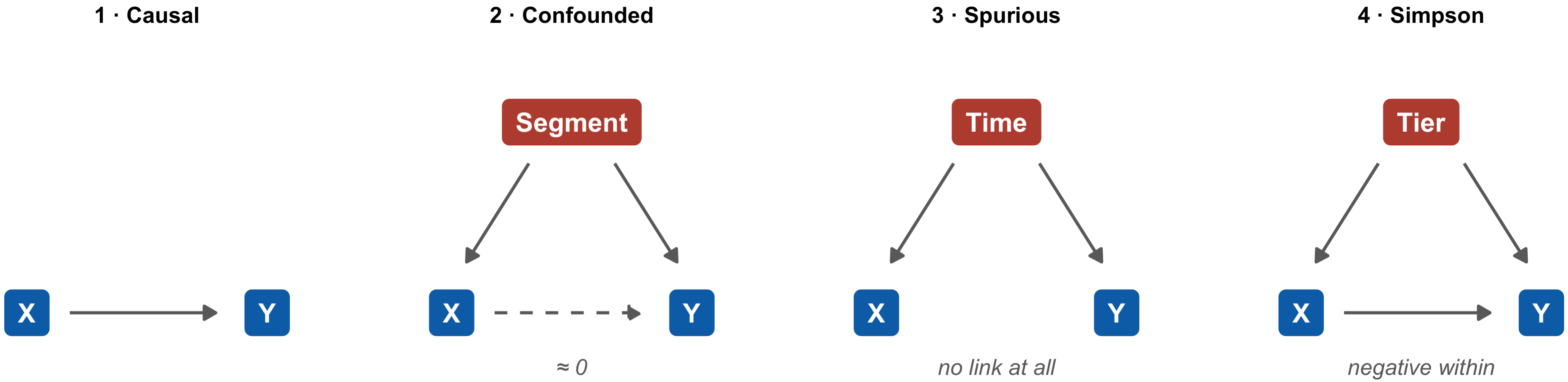
The confounder is the customer's **underlying brand interest**: it makes people **subscribe** *and* it makes them **buy**. Randomizing delivery — panel (a) — cuts its left arrow, and only the **+3** remains (more on this later).

Four ways to get the same scatter plot



- All four panels show a similar positive correlation — **the data alone cannot distinguish them**
- **Panel 2 — confounded:** a third variable creates the relationship (color = customer segment: loyal customers get more newsletters *and* spend more) — **within each segment, it vanishes**
- **Panel 3 — spurious:** any two trending series correlate — *Internet Explorer vs. murders*; time is the ultimate confounder
- **Panel 4 — Simpson's paradox:** confounding at full strength — *price vs. units across product tiers:* pooled, the slope is positive; **within every tier, it's negative.** Aggregation can *reverse* the truth, not just inflate it

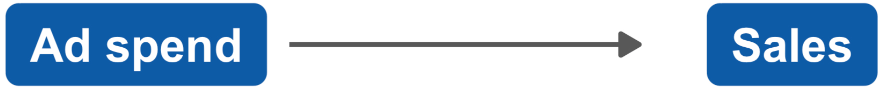
The four data-generating processes

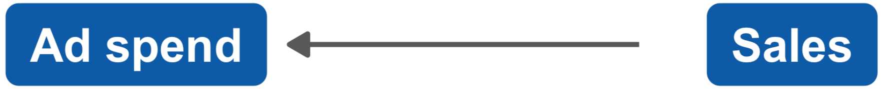

1 · Causal
2 · Confounded
3 · Spurious
4 · Simpson
Segment
Time
Tier
X
Y
X
Y
X
Y
X
Y
≈ 0
no link at all
negative within

> Four similar scatter plots — four different worlds. The plot shows the **correlation**; only the **DGP** says what X does to Y — and thus what happens if you **act on X**.

Reverse causality: who is causing whom?

Observed: regions with higher ad spend show higher sales.





Story 1: advertising drives sales.

Story 2: budgets are **set as a % of sales** — sales drive ad spend.

> Both stories produce identical data. Marketing budget rules make reverse causality a **routine** hazard in marketing data, not an exotic one.

The classical conditions for causality

For X to be accepted as a **cause** of Y, three conditions must hold:

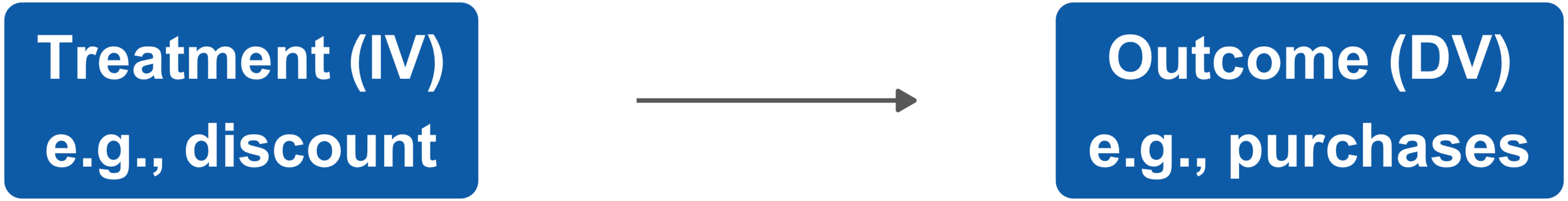
1. **Concomitant variation** — X and Y vary together as the hypothesis predicts. *The data establish this directly — all four scatter panels satisfy it. It is the cheap condition.*
2. **Temporal order** — the cause precedes the effect. *Sometimes checkable in data — but last slide's ad-budget example shows how it fails: sales also drive spending.*
3. **Absence of other possible causal factors** — no alternative explanation accounts for the association. *No dataset can certify this on its own: intent, segments, time, tiers…*

> The entire difficulty of causal inference is **condition 3** — and this course teaches **strategies for making it plausible**.

Classical formulation after Field, Miles & Field (2012) — see the course book, §1.4.

1.5 Variables and their roles

The two lead roles

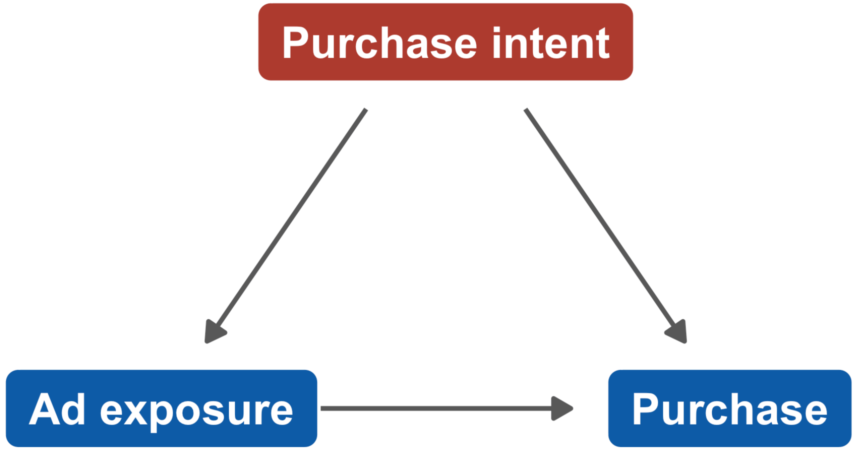


Treatment (IV)
e.g., discount

Outcome (DV)
e.g., purchases

- **Dependent variable (DV)** — the *outcome*: the quantity the manager cares about (purchases, churn, willingness to pay)
- **Independent variable (IV)** — the *presumed cause*: the variable whose influence we want to establish (discount, ad exposure, price)
- In **experiments**, the independent variable under the researcher's control is called the **treatment**; its values define the *experimental conditions* (treatment vs. control group)

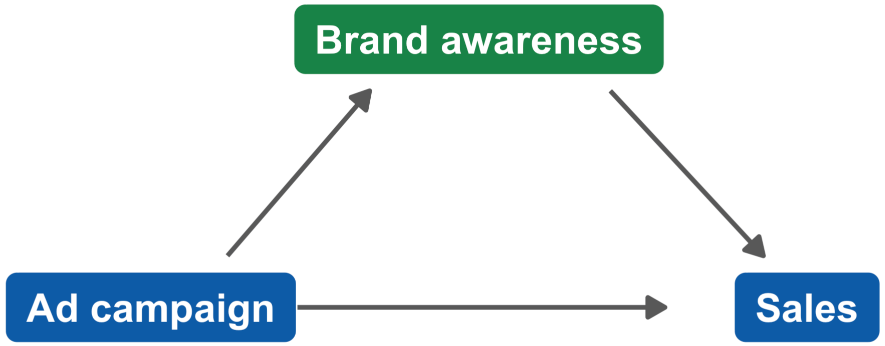
The villain: the confounder



- Causes **both** treatment and outcome
- Creates correlation **without** causation — panel 2 of our gallery
- **The eBay confounder:** purchase intent → people search for “eBay” → see the ad → and buy anyway
- The *Observational data* sessions build the toolkit for dealing with confounders

> When the confounding arises because units **choose their own exposure** — customers pick the channel, subscribe to the newsletter, install the app, join the loyalty program — the resulting distortion is called **selection bias**. In marketing data, it is the default, not the exception: *almost everything customers do, they chose to do.*

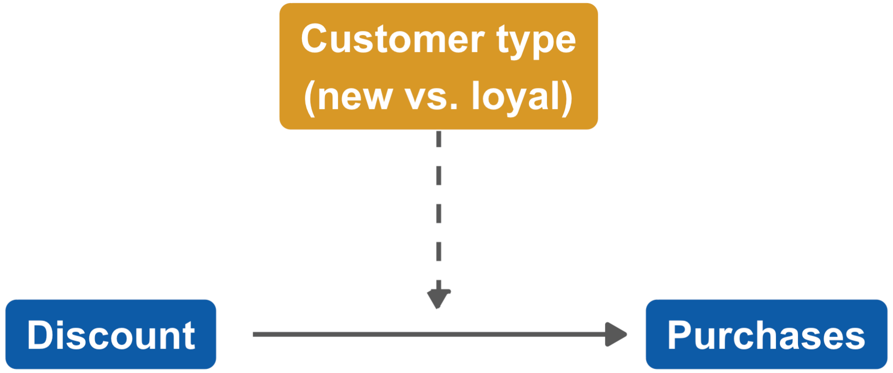
Supporting roles: mediator and moderator



Brand awareness

Ad campaign

Sales



Customer type (new vs. loyal)

Discount

Purchases

Mediator — the *how*

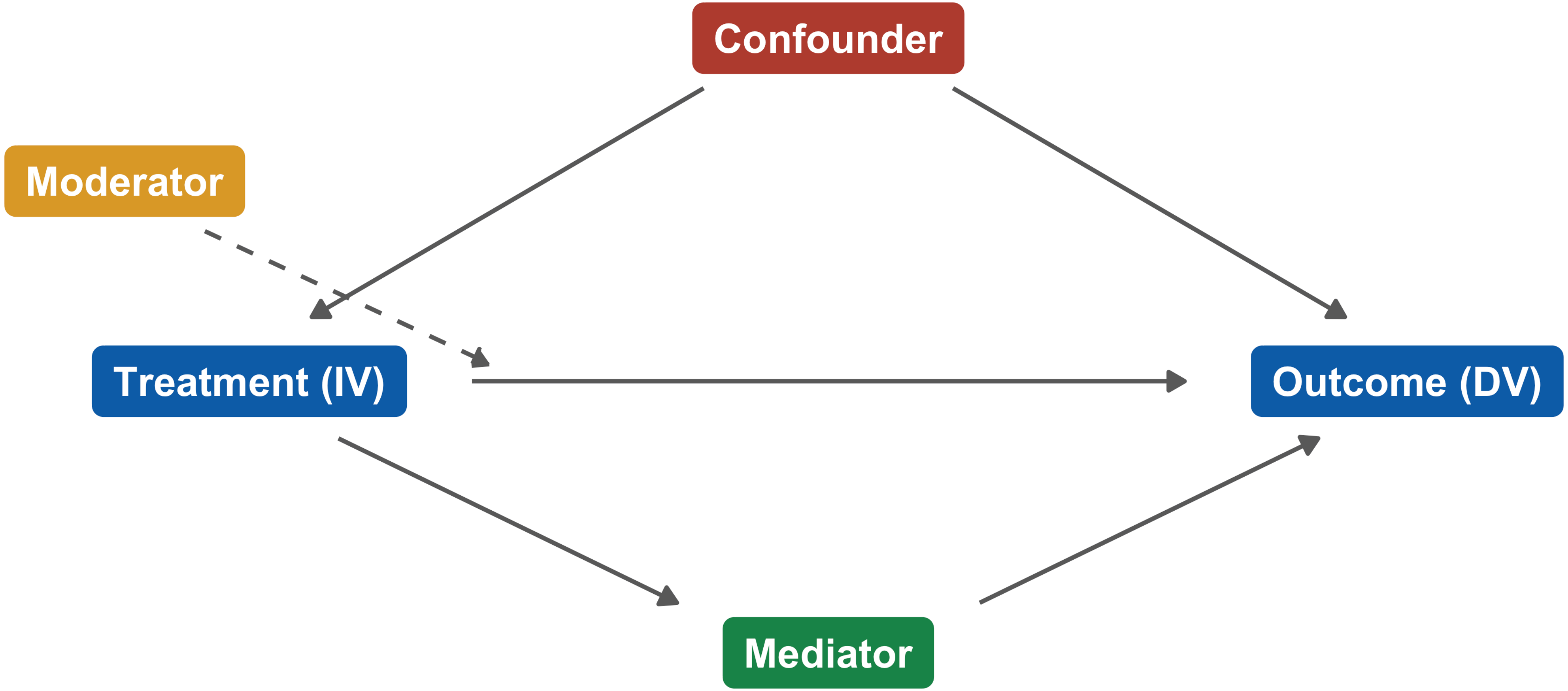
Lies **on the path**: the mechanism through which X works.
It tells you **what to optimize** (creative that builds awareness).

Moderator — the *for whom / when*

Points **at the arrow**: changes the **strength** of the effect.
“The discount works for new customers, not loyal ones.”

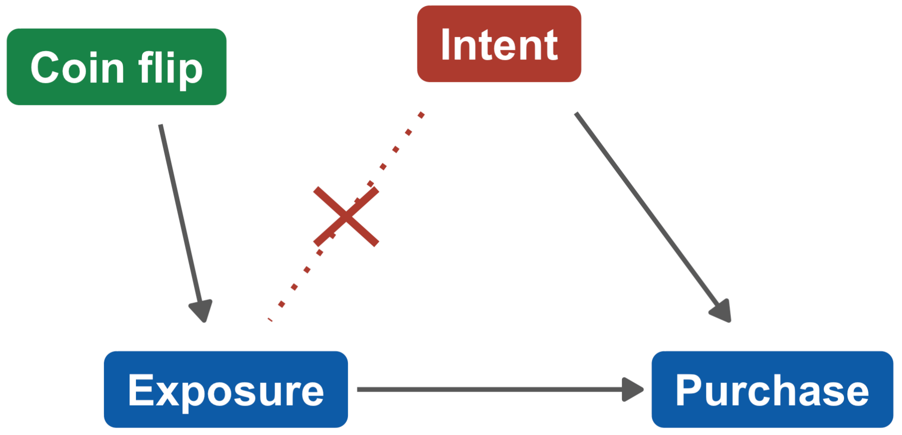
> Today: **recognize** the roles. Chapter 6 teaches how to **handle** each one — and shows that mishandling them (e.g., “controlling” for a mediator) can change the conclusions from an analysis.

The full cast around one causal question


Confounder
Moderator
Treatment (IV)
Outcome (DV)
Mediator

1.6 Randomization

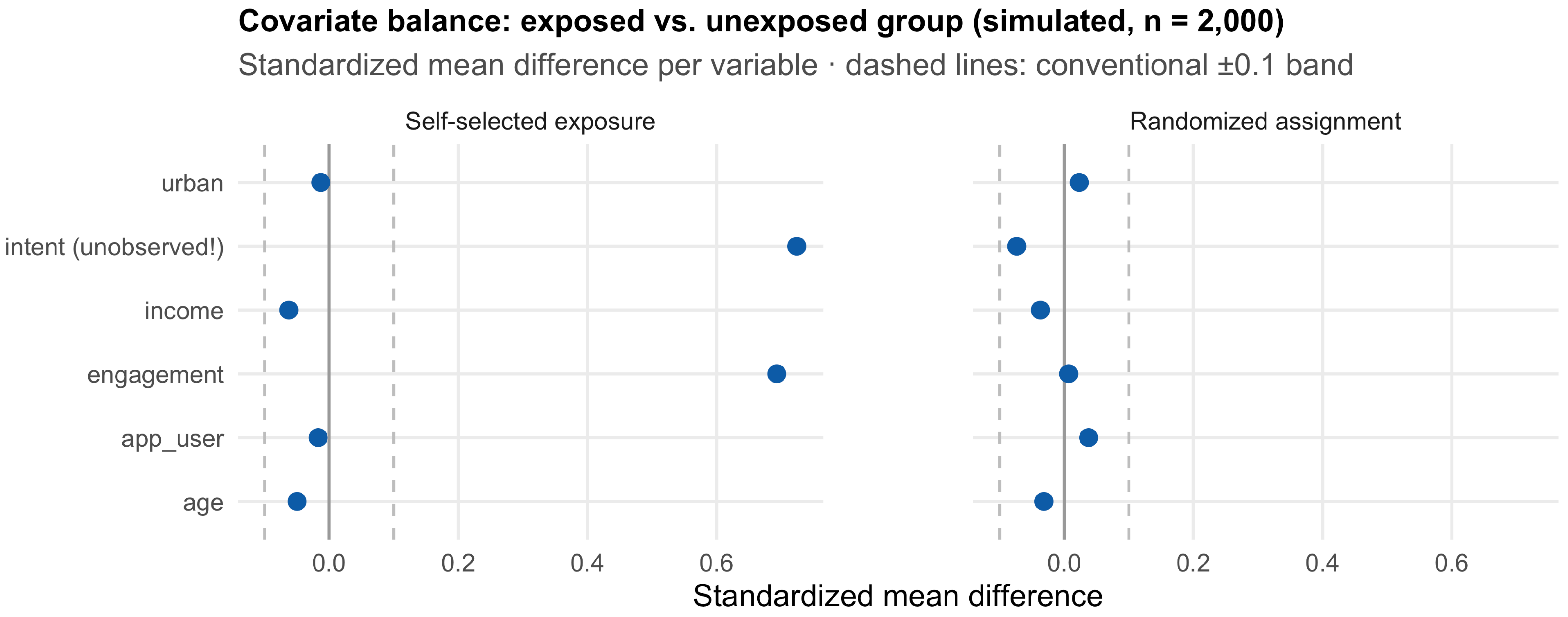
Randomization cuts every arrow into the treatment



- Assignment depends **only on the coin** — not on intent, engagement, income, mood, …
- **Systematic variation** — differences in the outcome created **by the treatment**: this is what we want to measure
- **Unsystematic variation** — differences from everything else (age, mood, time of day, …): randomization spreads it **evenly across groups**
- So group differences = **treatment effect + noise**. No bias — and the noise is quantified in Module 3

> Randomization does **not remove** other influences on the outcome — it makes them **unsystematic**: noise instead of bias. Noise averages out; bias does not.

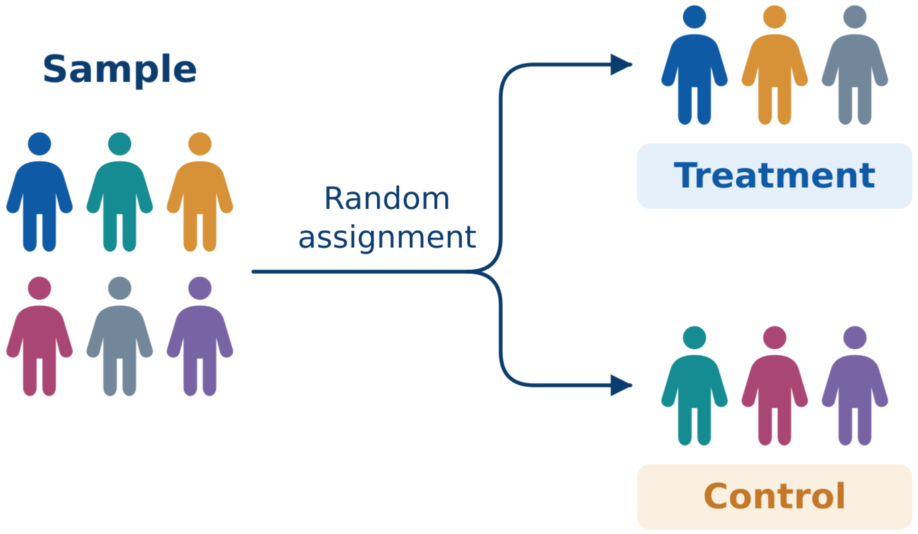
“Balanced — even on what we never measured”



> Left: the groups differ systematically — including on **intent**, which we (realistically) never observe. Right: one coin flip balances everything at once, **without knowing what needs balancing**.

Two ways to randomize

Between-subjects



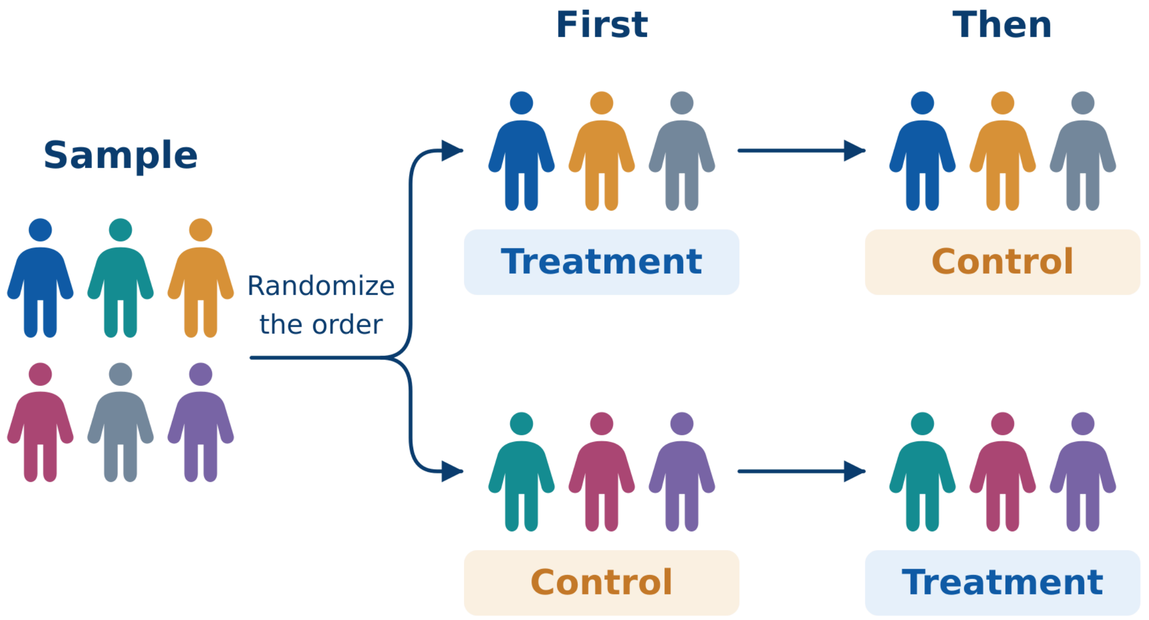
Sample

Random assignment

Treatment

Control

Within-subjects



First

Then

Sample

Randomize the order

Treatment

Control

Control

Treatment

Each person is randomly assigned to **one** condition — we compare **different people**.

Each person experiences **all** conditions, in random order — we compare **the same people**.

Between vs. within subjects: pros and cons

	Between-subjects	Within-subjects
Design	Each person sees **one** condition	Each person sees **all** conditions (repeated measures)
Example	Group A: targeted ad · Group B: generic ad	Everyone rates ad no. 1, then ad no. 2
Comparison	**Across people:** mean(A) vs. mean(B)	**Within person:** condition 1 vs. condition 2
Pros	No order or carryover effects · **realistic:** customers usually see only one version	Each person is their **own control** → can require **fewer participants**, *if* responses are correlated and carryover is limited
Cons	**More participants** needed · groups differ by chance → more noise	**Carryover:** the first condition can change the response to the second (learning, anchoring, a purchase already made) — counterbalancing helps, but cannot undo it

> **Rule of thumb:** within-subjects for **judgments** that do not change the person (rating ads, tasting products). Between-subjects for **behavior** (purchases, clicks) — or whenever the first exposure could change the response to the second. Online A/B tests keep each user in **one** condition.

Why the industry runs on experiments

> *“Our success at Amazon is a function of how many experiments we do per year, per month, per week, per day.”*
>
> — Jeff Bezos

> *“At any given point in time, there isn’t just one version of Facebook running, there are probably 10,000.”*
>
> — Mark Zuckerberg

A/B test = industry name for a between-subjects randomized experiment, run continuously and at scale.

Guidelines for field experiments

1. **Choose the unit of randomization.** Finer units (users rather than regions) → more statistical power — but weigh against logistics and spillovers. In digital environments, user-level A/B tests make fine granularity cheap
2. **Watch for spillovers & crossover.** *Spillover*: Treated units can affect untreated ones (product adoption spreads to friends; a tipping prompt changes driver behavior for *all* riders — Uber, Chandar et al. 2019). If spillovers loom, randomize at a **higher level** (households, cities). *Crossover*: a unit meant for control gets treated anyway (multiple devices, shared accounts)
3. **Complete vs. stratified randomization.** Pure coin flips can leave small samples imbalanced by chance. **Stratify**: split units into homogeneous subgroups first, randomize within each — worth it when a covariate strongly predicts the outcome

Lambrecht & Tucker (2016); Chandar, Gneezy, List & Muir (2019).

The price of observational data

Experimental research

> *“To find out what happens when you change something, it is necessary to change it.”*
>
> — Box, Hunter & Hunter (1978)

Observational research

> *“Behind any causal conclusion there must be some causal assumption, untested in observational studies.”*
>
> — Pearl (2009)

> That untested assumption has a name: **no unobserved confounders** — every variable driving both treatment and outcome has been measured and correctly adjusted for.

- **Untestable in principle**: you cannot check balance on variables you did not observe
- Experiments **buy** the assumption *by design*; observational studies must **argue** for it — with domain knowledge, not statistics

In-class reading: does personalization pay?

HARVARD BUSINESS REVIEW · DECEMBER 2021

Does Personalized Advertising Work as Well as Tech Companies Claim?

Bart de Langhe & Stefano Puntoni

→ https://hbr.org/2021/12/does-personalized-advertising-work-as-well-as-tech-companies-claim

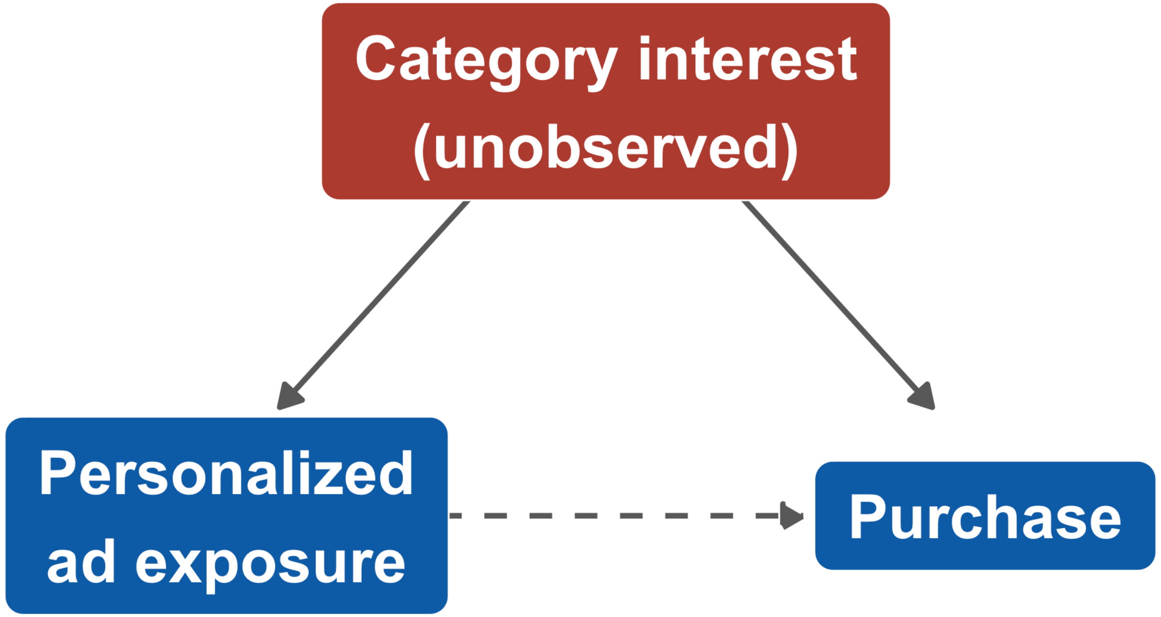
Then discuss:

1. According to its **objective**, what type of research question is being asked here — descriptive, predictive, causal, or prescriptive?
2. What exactly makes it **difficult to measure** the effect of personalized ads on sales?

Take ~10 minutes to read the article.

de Langhe & Puntoni (2021), “Does Personalized Advertising Work as Well as Tech Companies Claim?” *Harvard Business Review*, December 16.

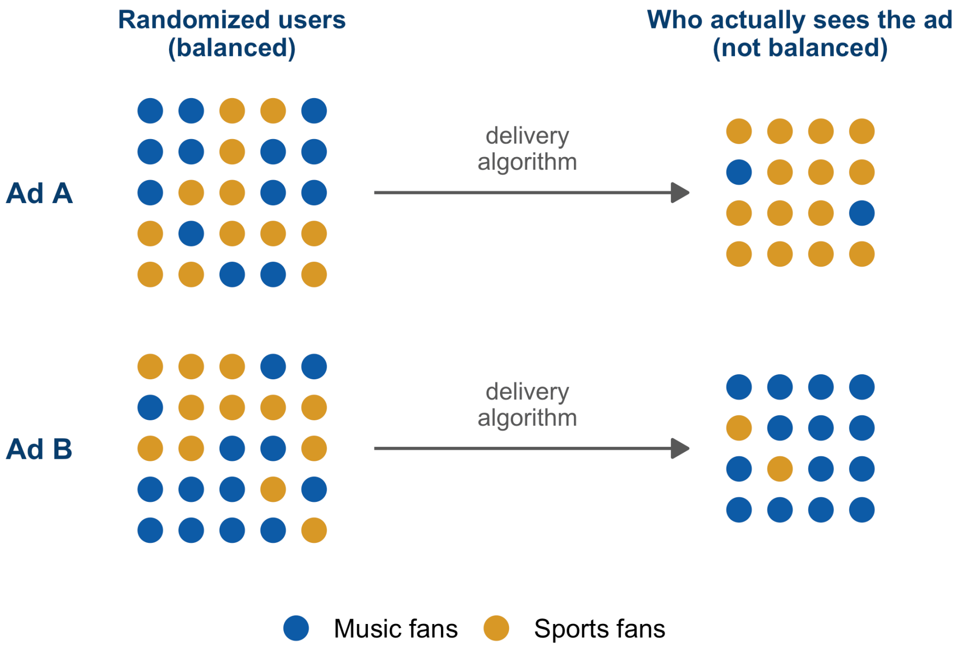
Threats to causal inference: who sees a personalized ad?


Category interest (unobserved)
Personalized ad exposure
Purchase

the lift the dashboard reports

Selection bias, machine-made: the targeting algorithm shows the ad *to people already interested in the category* — the platform **optimizes** for finding exactly the users who would buy anyway. Nobody chose anything; the selection happened inside the algorithm.

Even A/B tests can break: divergent delivery


Randomized users (balanced)
Who actually sees the ad (not balanced)
Ad A
delivery algorithm
Ad B
delivery algorithm
● Music fans
● Sports fans

- **Divergent delivery:** ad platforms' algorithms deliver different ads to **systematically different mixes of users — even during A/B tests**, confounding the effect of ad *content* with *targeting* effects
- **Consequence for inference:** the lack of balance can distort both the **magnitude and the sign** of estimated ad differences
- **Managerial implication:** platform A/B tools are useful for picking the better-performing ad **within the same platform** — they cannot reveal the true causal effect of ad content **across contexts, platforms, or audiences**

Braun & Schwartz (2025), "Where A/B Testing Goes Wrong: How Divergent Delivery Affects What Online Experiments Cannot (and Can) Tell You About How Customers Respond to Advertising", *Journal of Marketing*, 89(2), doi:10.1177/00222429241275886. Illustration: own, stylized.

1.7 Quasi-experiments

Between the poles: quasi-experiments

> *“Quasi-experimental tools mimic the random assignment that is inherent in lab experiments and that is often referred to as the ‘gold standard’ for identifying causal relationships.”*
>
> — Goldfarb & Tucker (2014)

- The “randomization” comes from an **exogenous shock** — a market entry, a policy change, a staggered rollout — **not controlled by the researcher**
- Units still **self-select** into treatment and control → the challenge is to argue that the shock **credibly proxies for random assignment**
- Often **fewer external-validity concerns** than lab experiments: real people, real markets, real money
- Workhorse estimator: **difference-in-differences (DiD)**

A spectrum of designs

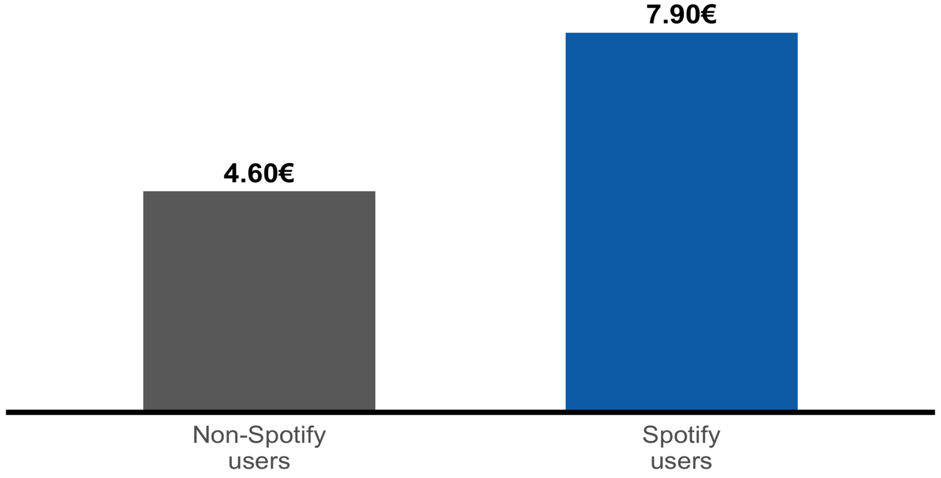
	Experiment	Quasi-experiment	Observational
Who assigns treatment?	The researcher (coin flip)	External event / rule	The units themselves
Example	A/B test of a discount	Platform launches in some countries first	Newsletter readers vs. non-readers
Key requirement	Correct execution	“As-good-as-random” timing/rule	**No unobserved confounders**

Example: does streaming cannibalize music sales?

- **Wlömert & Papies (2016):** effect of music streaming on (1) demand in **other channels** (downloads, CDs) and (2) **total** music industry revenues
- A typical **multi-channel distribution** problem in marketing
- **Spotify's market entry** = quasi-experiment
- Observation: Spotify users spend **more** on music in other channels than non-users

Monthly spending on CDs & downloads

One point in time



What the press concluded:

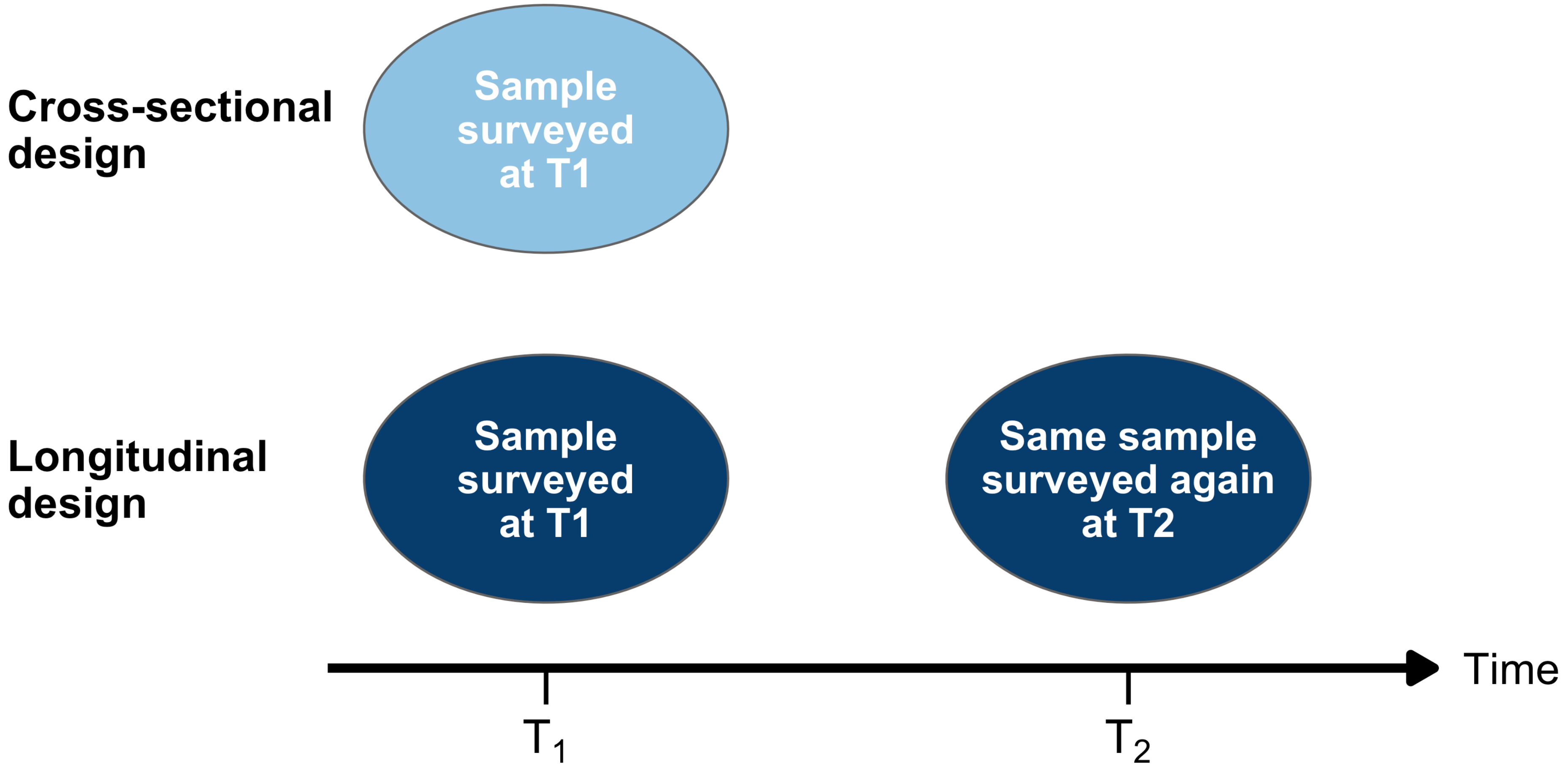
> *"Netflix Isn't Killing Movie Theaters, Study Shows"*
> — Variety, 2018

> *"Spotify Users Are Twice as Likely to Purchase a Download…"*
> — Digital Music News, 2012

Is this evidence of a causal effect? Would you conclude that streaming *boosts* CD and download sales?

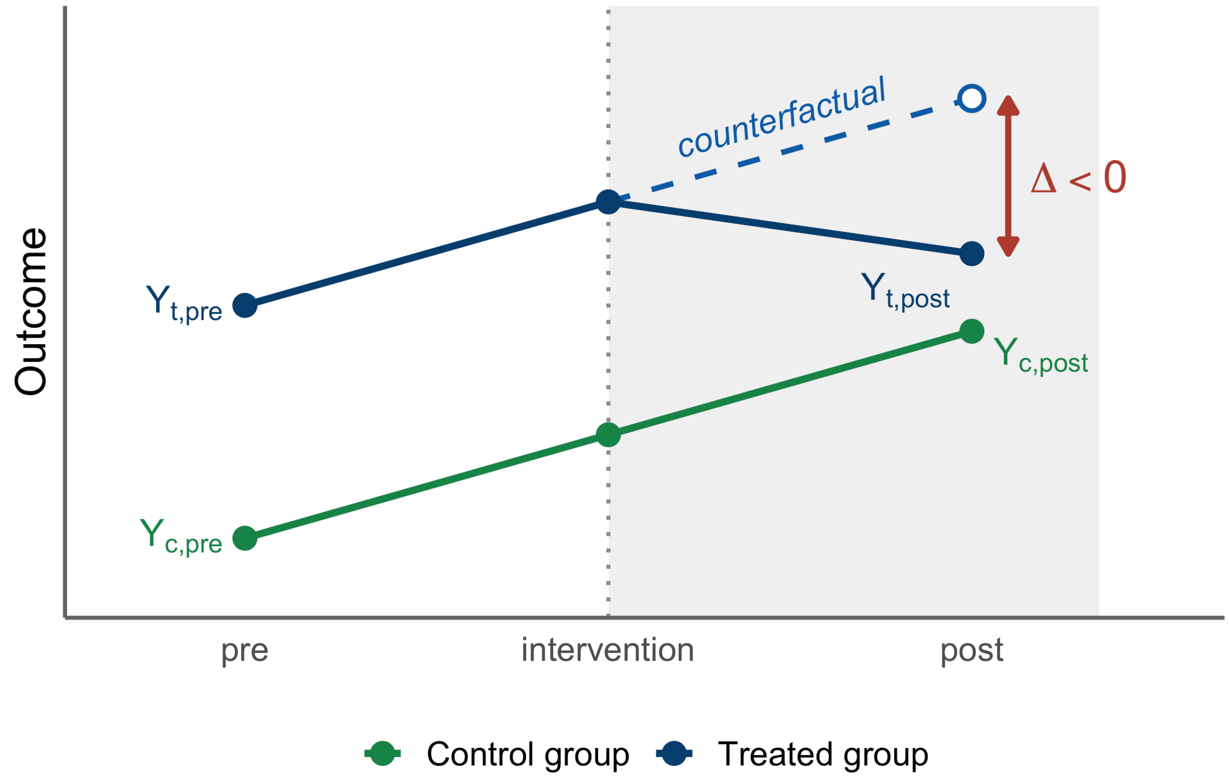
Wlömert & Papies (2016), *International Journal of Research in Marketing*, 33(2), 314–327. Press headlines: Variety (2018); Digital Music News (2012).

Snapshots vs. following people over time


Cross-sectional design
Sample surveyed at T1
Longitudinal design
Sample surveyed at T1
Same sample surveyed again at T2
Time
T_1
T_2

A **longitudinal (panel) design** observes the **same units** repeatedly — so we can see how each group **changes** over time, not just where it stands.

Difference-in-differences: the logic



Period	Treated	Control	Counterfactual
Before	$Y_{t,\text{pre}}$	$Y_{c,\text{pre}}$	$Y_{t,\text{pre}}$
After	$Y_{t,\text{post}}$	$Y_{c,\text{post}}$	$Y_{t,\text{pre}} + (Y_{c,\text{post}} - Y_{c,\text{pre}})$

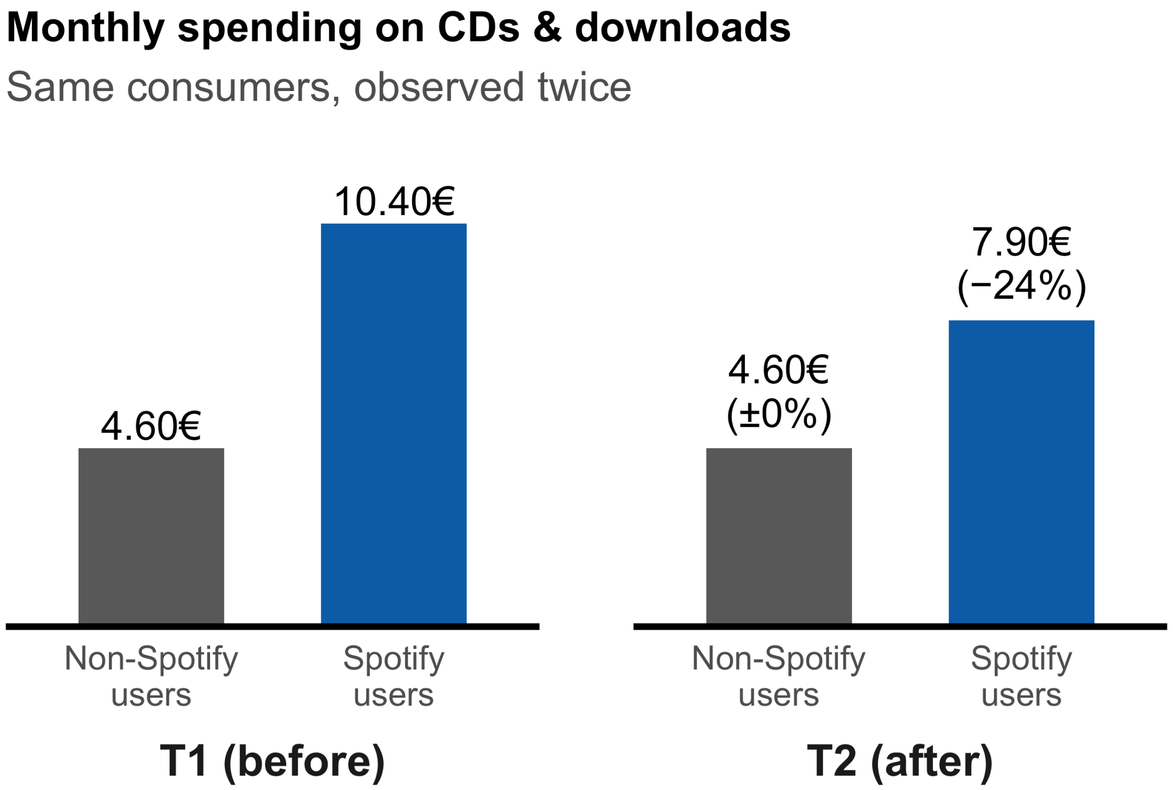
Effect of the treatment on the treated (Δ):

$$\Delta = (Y_{t,\text{post}} - Y_{t,\text{pre}}) - (Y_{c,\text{post}} - Y_{c,\text{pre}})$$

> **Key assumption — parallel trends:** without the treatment, the treated group would have changed **by the same amount** as the control group.

Adapted from Varian (2016), "Causal inference in economics and marketing", *PNAS*, 113(27), 7310–7315.

DiD in action: the Spotify example



Period	Treated	Control	Counterfactual
Before	10.40€	4.60€	10.40€
After	7.90€	4.60€	10.40€ + (4.60€ − 4.60€)

Effect of the treatment on the treated (Δ):

(7.90€ − 10.40€) − (4.60€ − 4.60€) = **−2.50€**

> **Same data, opposite conclusion:** the snapshot says Spotify users spend **+3.30€ more**; following the same people over time says Spotify **reduces** their spending by **2.50€** per month. The snapshot measured *who adopts*, not *what adoption does*.

Wlömert & Papies (2016), *International Journal of Research in Marketing*, 33(2), 314–327. Stylized figures.

1.8 From business problem to research design

The translation template

Manager: *“I want to know whether our loyalty program is worth the money.”*

Design element	Your translation
Outcome (DV)	Customer contribution margin, 12 months
Treatment (IV)	Loyalty program membership
Unit of analysis	Customer
Question type	Causal (→ prescriptive decision)
Likely confounders	Purchase frequency & engagement *before* joining
Feasible design?	Randomize invitations? Staggered regional rollout? Observational + adjustment?

> A precise, falsifiable formulation (“membership increases 12-month margin”) beats a vague aspiration (“the program creates value”) — it dictates data, design, and analysis.

Your turn (pairs, 10 min)

Translate one of these into the template:

1. “Should we invest more in influencer marketing?”
2. “Is our app driving retention, or do loyal customers just download the app?”
3. “Did last spring’s rebranding damage sales in Germany?”

For each: DV, IV, unit, question type, two plausible confounders, and the **best feasible design** (experiment / quasi-experiment / observational).

1.9 The analytics workflow

Where AI helps — and where it cannot replace you

Workflow step	Nature of the work	AI assistant…
Formulate the question	**Judgment**	can challenge you — cannot know your business
Choose design & method	**Judgment**	suggests options — *you* own the assumptions
Write & debug R code	Mechanical	**excellent** — use it freely, including in this course
Interpret results	**Judgment**	drafts text — but cannot verify the causal claim
Communicate to stakeholders	Mixed	strong support, needs your framing

> **Course policy in one line:** use AI to *accelerate* your learning of the judgment steps, not to *skip* them. Assessments in this course are designed so that the judgment steps are what earns the points.

One map for the whole course

Module	Sessions	Driving question
1 · Questions, decisions & data	1 *(today)*	Which question am I really asking?
2 · Describing data	2	What is going on?
3 · Uncertainty	3	How sure can I be?
4 · Learning from experiments	3 / 4	Did it work?
5 · Learning from observational data	5 / 6	What if we cannot randomize?
6 · Mechanisms & heterogeneity	7	Why — and for whom — does it work?
Survey design	8	How do I build a study that can answer my question?
Group project coaching	2 online sessions	Your own experiment, end to end
Final exam (in person)	Tue, Dec 15	—

Learning check — can you spot the reasoning?

1. An analyst reports: *“App users have 2× the lifetime value of non-users — the app clearly pays off.”* Which question type is being **answered**, which is being **claimed**? Name the most plausible confounder.
2. A platform’s advertising dashboard compares purchases of users who clicked an ad vs. users who saw it and didn’t click. Is this an experiment? What role does “interest in the product” play?
3. Your CMO ran a 4-week price promotion **in Vienna only**. Weekly sales in Vienna rose from **200k€ to 230k€**; in Graz (no promotion) from **150k€ to 160k€** over the same weeks. What is the **difference-in-differences** estimate? What must you **assume** for it to be causal?
4. Confounder, mediator, or moderator? (a) The newsletter increases purchases *because it increases site visits*. (b) The discount works much better *for students*. (c) Customers who buy our cereal also buy more of our toys — *households with more children* buy more of both.

Learning check — true or false?

1. A sufficiently large dataset removes the need for the no-unobserved-confounders assumption.
2. Randomization balances treatment groups even on characteristics that were never measured.
3. A variable can be an excellent predictor of an outcome without having any causal effect on it.
4. Two time series that both trend upward will be correlated — even if they are completely unrelated.
5. If Spotify users spent more on music than non-users *before* adopting Spotify, difference-in-differences cannot be used.
6. In a platform A/B test, randomizing users into ad A vs. ad B guarantees that both ads reach comparable audiences.

> **Preview:** the paper quiz at the start of each session looks like this — short, conceptual, no R.

Next session

Module 2 · Describing data — *What is going on?*

- Getting data into R, cleaning it, and not trusting it too early
- Summary statistics and the visualizations that earn their place
- **Before the session:** complete the R onboarding module — link on the course site.

Course book chapter 1 (consolidation reading for today) is online — including all simulated examples from these slides, with their code.